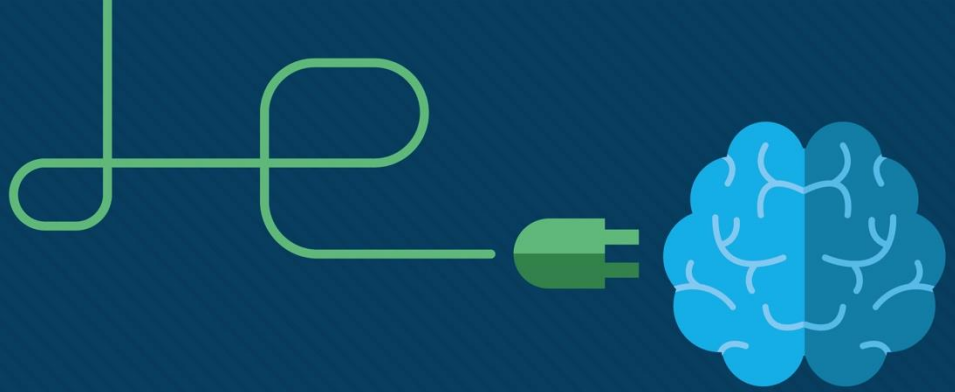

Module 5
Wireless Networks

ITNET03

Advanced Routing and Switching

Module Objectives

Module Title: Wireless Networks

Module Objectives:

- Describe wireless LAN technologies, standards, and components.
- Describe wireless LAN architectures and operation
- Describe wireless LAN threats and security measures
- Configure wireless LANs for SOHO networks
- Configure wireless LANs for enterprise networks

Module References:

- CCNAv7 SRWE – Module 12, 13

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5.1 Introduction to Wireless

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Mobility Support in an Organization

- Recent trends in work responsibilities and habits have people requiring mobility
  - People are not constrained to a fixed workspace
  - People still need to be connected to the network while moving
- Benefits of supporting mobility
  - Flexibility in adapting to rapidly changing needs and technologies.
  - Increased productivity by enabling people to be connected while on the move
  - Reduced equipment costs due to less cabling

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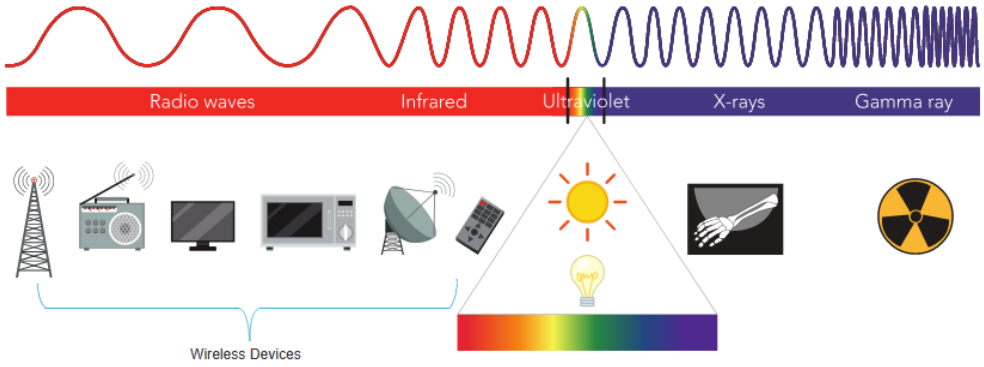
Characteristics of Wireless Communications

- Wireless communication technologies use radio signals to transmit data
  - Wireless signals are waves of electromagnetic energy travelling through the air
  - All wireless devices operate in the electromagnetic spectrum and transmit data at a specific range of **frequencies** (rate at which energy vibrates to form a signal).



- Wireless is considered **shared** medium
  - Affected other radio devices and electrical equipment that emit energy or communicate wirelessly at the same frequencies
  - Communicating devices take turns transmitting to avoid interference
  - Can be regulated by government authorities

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Types of Wireless Networks

- **Wireless Personal-Area Network (WPAN)**
  - Low power and short-range (20-30ft or 6-9 meters) network usually involving devices of a single user
  - Based on IEEE 802.15 standard and commonly uses the 2.4 GHz frequency.
  - e.g. Bluetooth, ZigBee

- **Wireless Local Area Network (WLAN)**
  - Medium sized networks up to about 300 feet usually involving devices within a single organization
  - e.g. IEEE 802.11

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Types of Wireless Networks

- **Wireless Metropolitan Area Network (WMAN)**
  - Large geographic area spanning an entire city or district.
  - Typically owned by an ISP
  - Uses specific licensed frequencies.
  - e.g. WiMAX (IEEE 802.16)

- **Wireless Wide Area Network (WWAN)**
  - Extensive geographic area for national or global communication.
  - Typically owned by an ISP
  - Uses specific licensed frequencies.
  - e.g. Cellular broadband (3G/4G/5G)

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802.11 Standards

802.11 WLAN standards define how radio frequencies are used for wireless links.

Standard	Frequency	Max Bandwidth	Characteristics
802.11a	5 GHz	54 Mbps	High speed, less prone to interference but shorter range and more easily obstructed by walls Prohibited in some countries
802.11b	2.4 GHz	11 Mbps	Low cost, good range, not easily obstructed More prone to interference from wireless devices
802.11g	2.4 GHz	54 Mbps	Fast, Low cost, good range, not easily obstructed More prone to interference from wireless devices Backward compatible with 802.11b
802.11n	2.4 and 5 GHz	150- 600 Mbps	Long range, high bandwidth. Backward compatible with a/b/g/n Uses MIMO technology hence requires multiple antennas
802.11ac	5 GHz	450 Mbps - 1.3 Gbps	Long range, high bandwidth, high client to access point ratio Uses MIMO technology with support for up to 8 antennas
802.11ax	2.4 and 5 GHz	3.5 – 14 Gbps	High-Efficiency Wireless (HEW) aka WiFi6 Limited devices with this capability for now Designed for high-density public environments

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Wireless Standards Organizations

Standards ensure interoperability between devices that are made by different manufacturers. Internationally, the three organizations influencing WLAN standards:

- **International Telecommunication Union (ITU)** – Regulates the allocation of radio spectrum and satellite orbits.
- **Institute of Electrical and Electronics Engineers (IEEE)** – Specifies how a radio frequency is modulated to carry information. Maintains the standards for local and metropolitan area networks (MAN) with the IEEE 802 LAN/MAN family of standards.
- **Wi-Fi Alliance** – Promotes the growth and acceptance of WLANs. It is an association of vendors whose objective is to improve the interoperability of products that are based on the 802.11 standard

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Wireless LAN Devices

- **Wireless NIC / Adapters**
  - Incorporate a radio transmitter/receiver used by devices to connect to a wireless network
  - Can be built-in or sold as an expansion card or USB device

- **Wireless Access Points (WAP)**
  - Wireless Access Point (AP)
  - Connects wireless end devices to a WLAN
  - Works like a ‘wireless switch’
  - May be classified as:
    - Autonomous - Standalone devices configured individually through a command line interface or GUI
    - Controller-based - automatically configured and centrally managed by a wireless controller.

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Wireless LAN Devices

- **Wireless Home Router**
  - All-in-one device commonly found in home or SOHO deployments
  - Combines the functions of
    - Access point – To provide wires access
    - Switch – To interconnect wired devices
    - Router - To provide a default gateway to other networks and the Internet

- **Wireless Controller (WLC)**
  - Used for centralized management of multiple access points in large deployments
  - A single configuration may be pushed to multiple access points at the same time
  - May be implemented as standalone devices, router modules or software

Cisco Virtual Controller

Cisco Wireless Controller on the Cisco Services Ready Engine (SRE)

Cisco Wireless Controller on the Cisco Services Ready Engine (SRE)

Wireless LAN Antennas

Wireless devices with external antennas may use different types depending on on the scenario where it is deployed:

- **Omnidirectional**
  - Provide 360-degree coverage.
  - Ideal in indoor areas such as houses and office areas to reach many users located in various directions from the antenna
- **Directional**
  - Focus the radio signal in a specific direction. Examples are the Yagi and parabolic dish.
  - Commonly found in outdoor deployments used to bridge networks across a distance wirelessly (e.g. between buildings)
- **Multiple Input Multiple Output (MIMO)**
  - Uses multiple antennas (Up to eight) to increase bandwidth by combining capacity.

5.2 WLAN Operation

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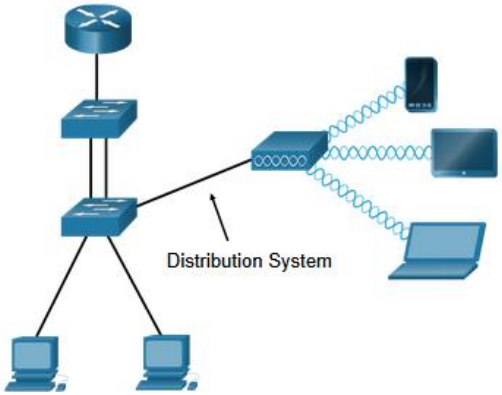
802.11 Wireless Topology Modes

Ad hoc Mode	Infrastructure Mode	Tethering
Connect clients in peer-to-peer manner without an AP.	Connects clients to the wireless network using an AP. 	An ad hoc topology wherein a smart phone or tablet with cellular data access is used to create a personal hotspot where wireless devices may connect.

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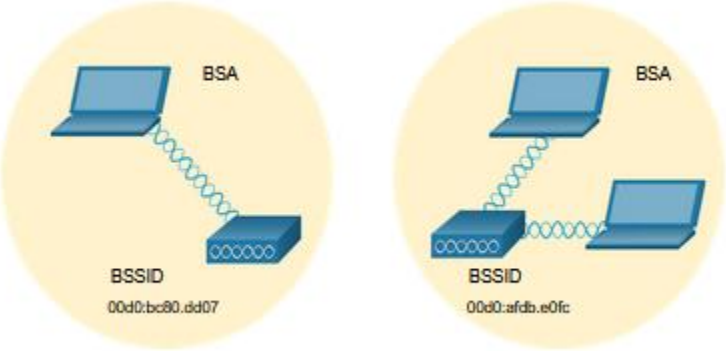
BSS and ESS

Infrastructure mode defines two topology blocks:

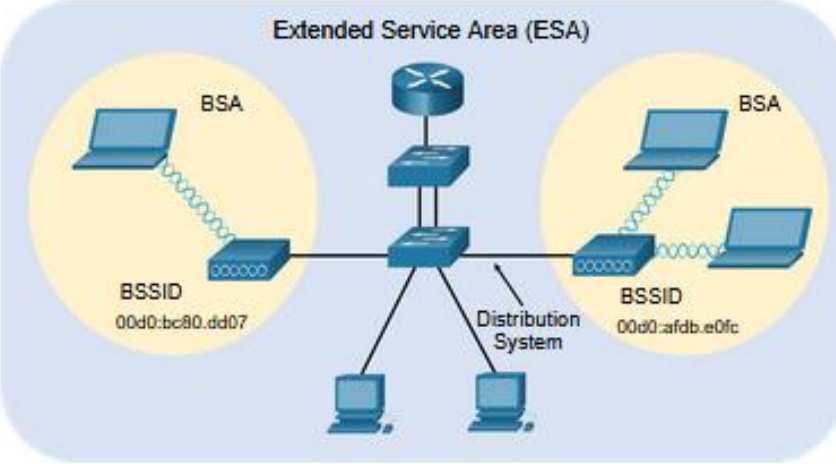
Basic Service Set

- Uses a single AP to connect all wireless clients.
- Each group of AP and its clients is called a basic service
- Commonly found in SOHO deployments where a single AP or router can provide the required network coverage area



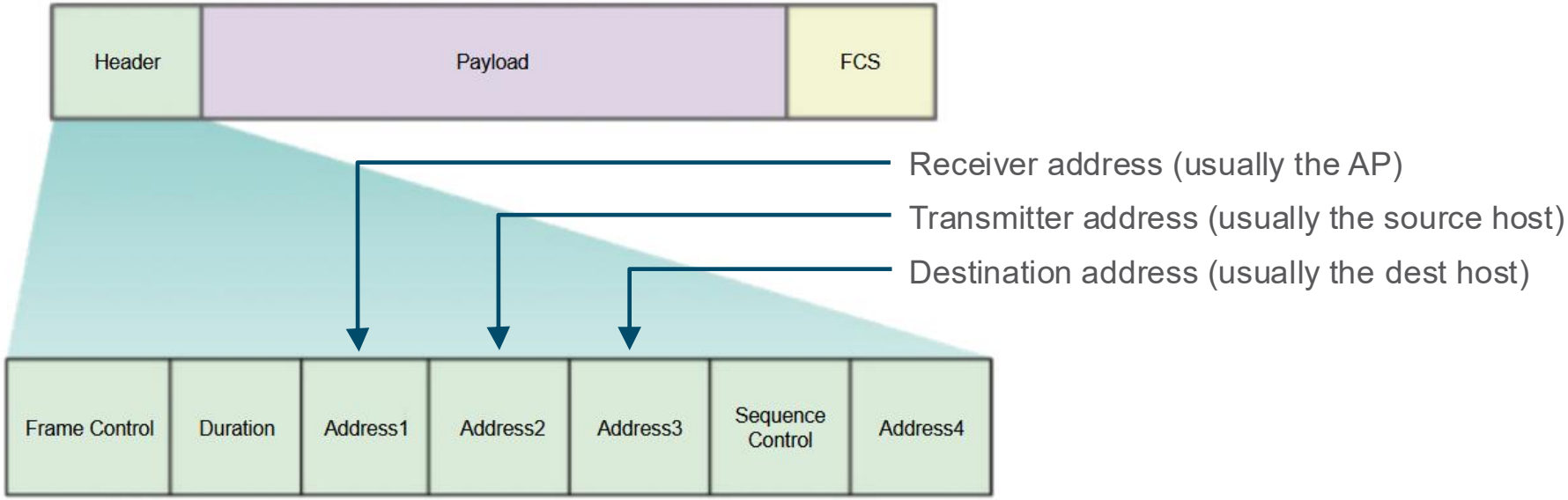
Extended Service Set

- A union of two or more BSSs interconnected by a wired distribution system .
- Clients in each BSS can communicate through the ESS.
- Commonly found in large deployments where several APs or routers are needed to cover the entire facility and connect devices to a single network



802.11 Frame Structure

The 802.11 frame format is similar to the Ethernet frame format, except that it contains more fields.



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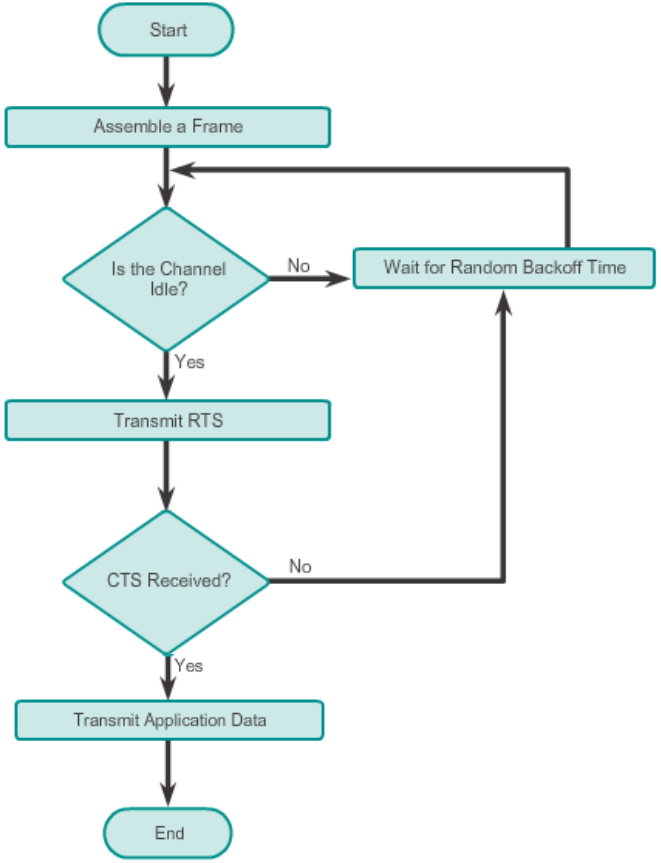
WLAN Operation
CSMA/CA

- WLANs are half-duplex and a client cannot “hear” while it is sending, making it impossible to detect a collision.
- WLANs use **carrier sense multiple access with collision avoidance (CSMA/CA)** to determine how and when to send data. A wireless client does the following:

1. Listen to the channel to see if it is idle, i.e. no other traffic currently on the channel.
2. Send a ready to send (RTS) message to the AP to request dedicated access to the network.
   - 2.1 If a clear to send (CTS) message is received from the AP granting access to send, transmit data
   - 2.2 If no CTS, wait a random amount of time before restarting the process.
3. Wait for acknowledgement
   - 3.1 If ACK received, transmission done,
   - 3.2 If no ACK receives, assume collision and restart the process



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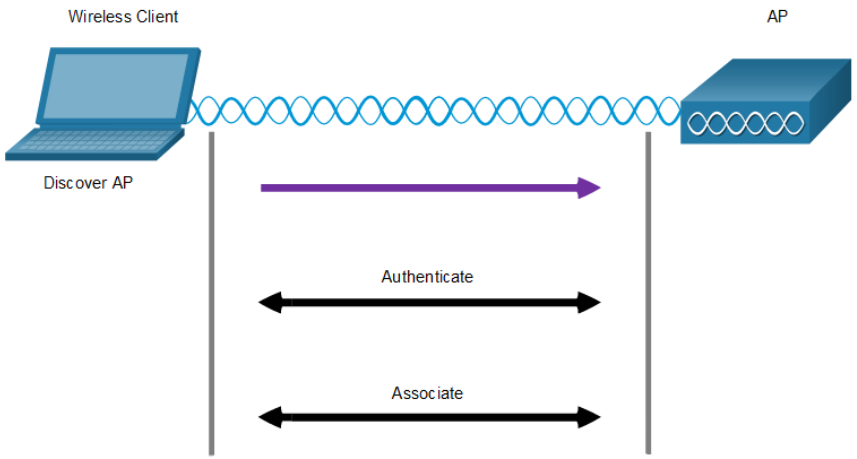
Wireless Client and AP Association



For wireless devices to communicate over a network, they must first associate with an AP or wireless router using a three stage process:

- Discover a wireless AP
- Authenticate with the AP
- Associate with the AP

To achieve successful association, a wireless client and an AP must agree on specific parameters:

- **SSID** – The client needs to know the name of the network to connect.
- **Password** – This is required for the client to authenticate to the AP.
- **Network mode** – The 802.11 standard in use.
- **Security mode** – The security parameter settings, i.e. WEP, WPA, or WPA2.
- **Channel settings** – The frequency bands in use.

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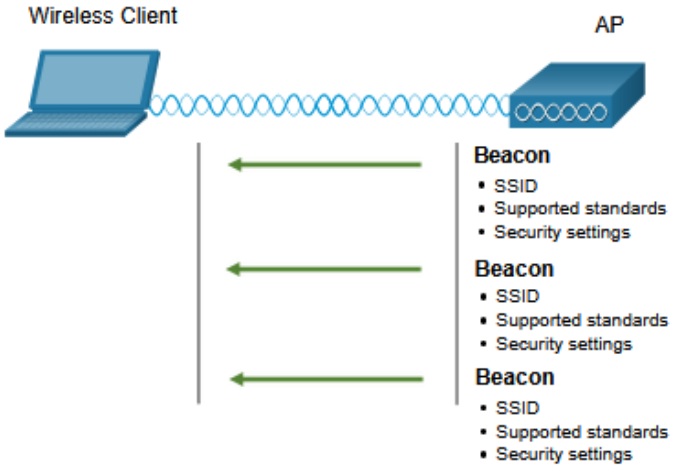
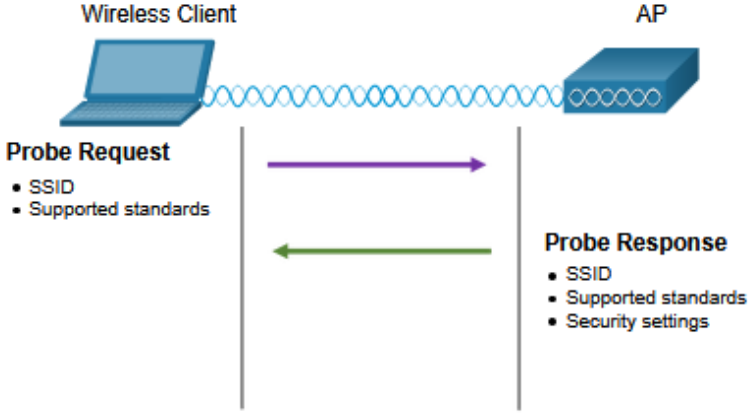
WLAN Discovery

Wireless clients connect to the AP using a passive or active scanning (probing) process.

- **Passive mode** – AP openly advertises its service by periodically sending broadcast beacon frames containing the SSID, supported standards, and security settings.



- **Active mode** – Wireless clients must know the name of the SSID. The wireless client initiates the process by broadcasting a probe request frame on multiple channels (hidden SSID network).

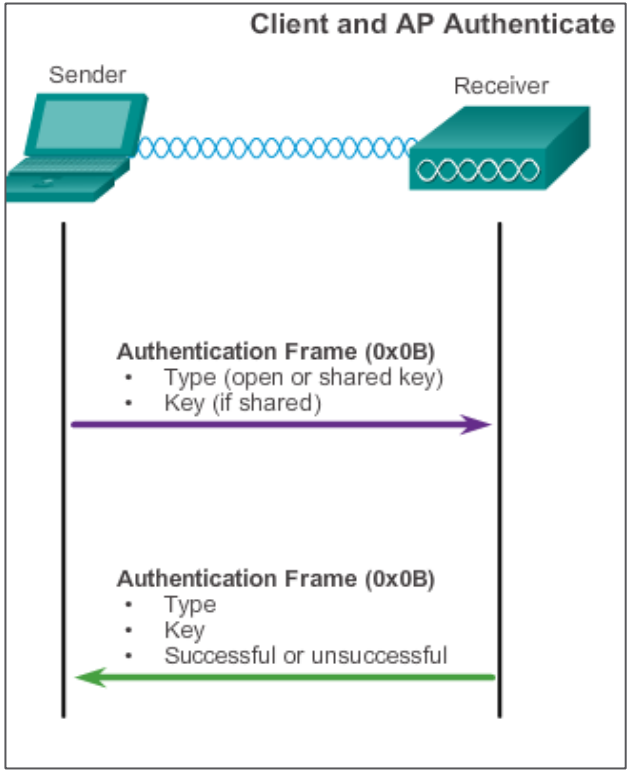


WLAN Authentication

Wireless networks may use 2 methods of authentication.

- **Open authentication –** A NULL authentication where the wireless client says “authenticate me” and the AP responds with “yes.” Used where security is of no concern.

- **Shared key authentication –** Technique is based on a key that is pre-shared between the client and the AP.



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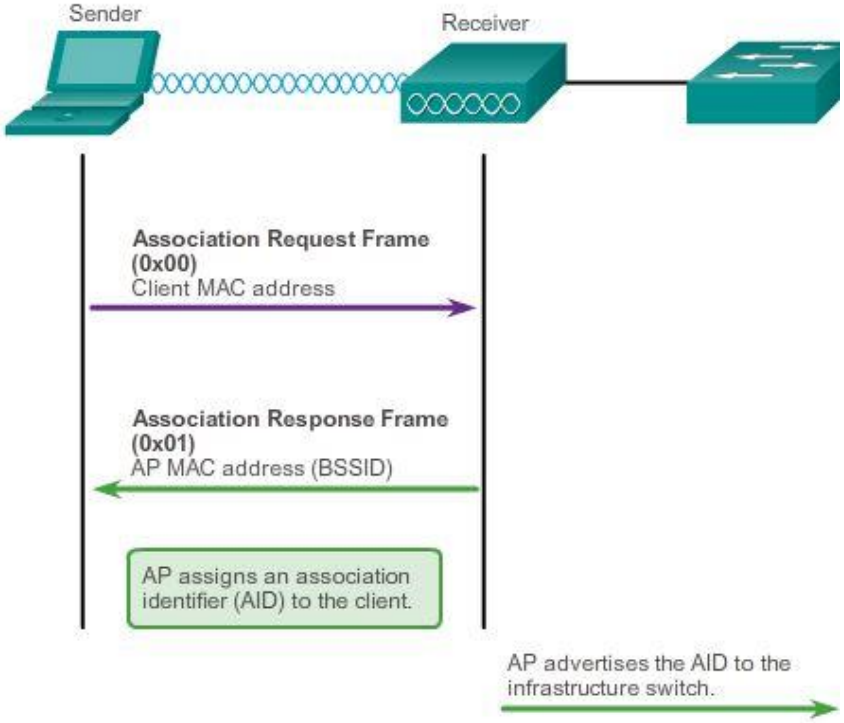
WLAN Association

- The association step finalizes the security and bit rate options for the link between client and AP.
- Establishes the data link between the WLAN client and the access point.
- The client learns the BSSID (MAC Address) of the access point.
- Access point maps a logical port called the association identifier (AID) to the WLAN client.



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5.3 WLAN Security

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Wireless Security Overview

- A WLAN is open to anyone within range of an AP and the appropriate credentials to associate to it.
- Attacks can be generated by outsiders, disgruntled employees, and even unintentionally by employees.
- Wireless networks are specifically susceptible to several threats, including the following:

Wireless intruders / freeloaders	Denial of Service (DoS) Attacks	Rogue APs	Interception of data

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DoS Attacks

Wireless DoS attacks can be the result of the following:

- Improperly configured devices
- A malicious user intentionally interfering with the wireless communication
  - Spoofed disconnect attack - Occurs when an attacker sends a series of “disassociate” commands to all wireless clients to cause all clients to disconnect from the WLAN.
  - CTS flood - The attacker repeatedly floods Clear to Send (CTS) frames so that all wireless clients sharing the RF medium receive the CTS assume that a host is given permission to transmit and so withhold transmissions
- Accidental interference
  - WLANs operate in the unlicensed frequency bands and are prone to interference from other wireless devices.
  - May occur from such devices as microwave ovens, cordless phones, baby monitors, and more

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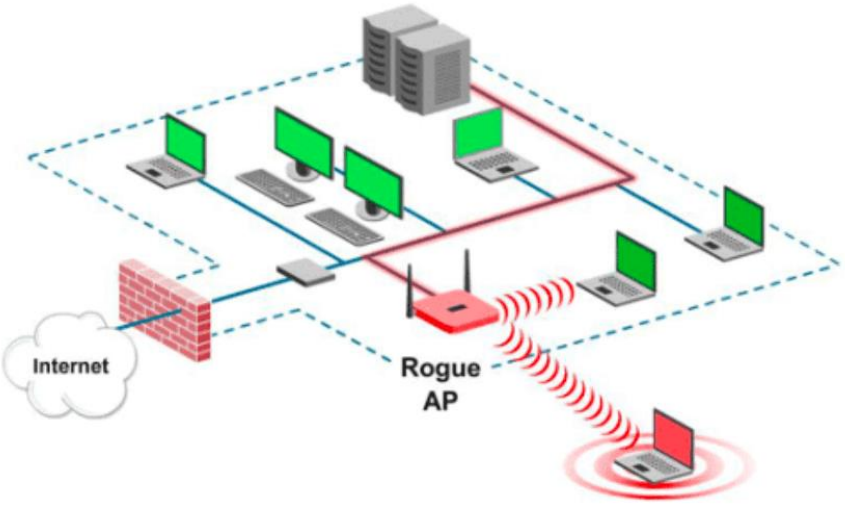
Rogue Access Points

- A rogue AP is an AP or wireless router that has been connected to a corporate network without explicit authorization and against corporate policy.
- Once connected, the rogue AP can be used by an attacker to capture MAC addresses, capture data packets, gain access to network resources, or launch a man-in-the-middle attack.
- To prevent the installation of rogue APs, organizations must configure WLCs with rogue AP policies and use monitoring software to actively monitor the radio spectrum for unauthorized APs.



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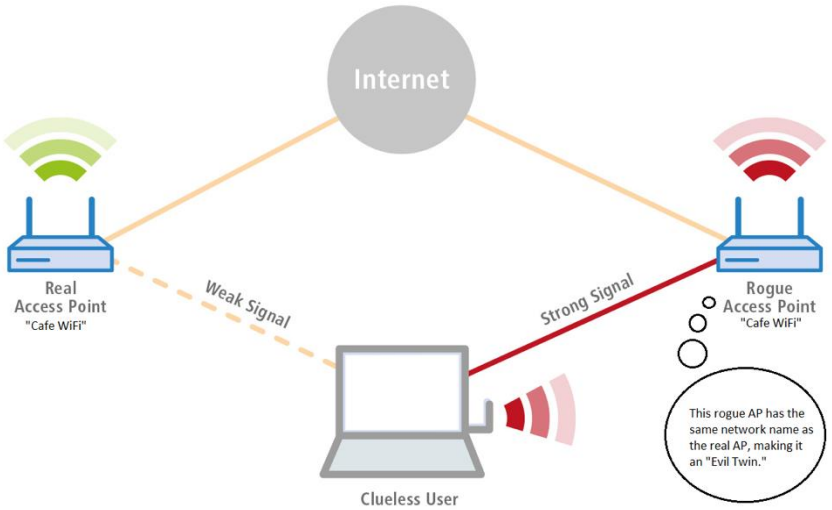
Man-in-the-Middle Attack

- In a man-in-the-middle (MITM) attack, the hacker is positioned in between two legitimate entities in order to read or modify the data that passes between the two parties.
- Evil Twin AP Attack
  - An attacker introduces a rogue AP and configures it with the same SSID as a legitimate AP in order to pretend to be part of the network.
  - Attacker can capture data of clients that connect to the rogue AP
  - Often done in public places offering free WiFi
- Defeating a MITM attack begins with identifying legitimate devices on the WLAN through authentication



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SSID Cloaking and MAC Address Filtering

To address the threats of keeping wireless intruders out and protecting data, two early security features were used and are still available on most routers and APs:

SSID Cloaking

- APs and some wireless routers allow the SSID beacon frame to be disabled. Wireless clients must be manually configured with the SSID to connect to the network.

MAC Address Filtering

- An administrator can manually permit or deny clients wireless access based on their physical MAC hardware address. In the figure, the router is configured to permit two MAC addresses. Devices with different MAC addresses will not be able to join the 2.4GHz WLAN.

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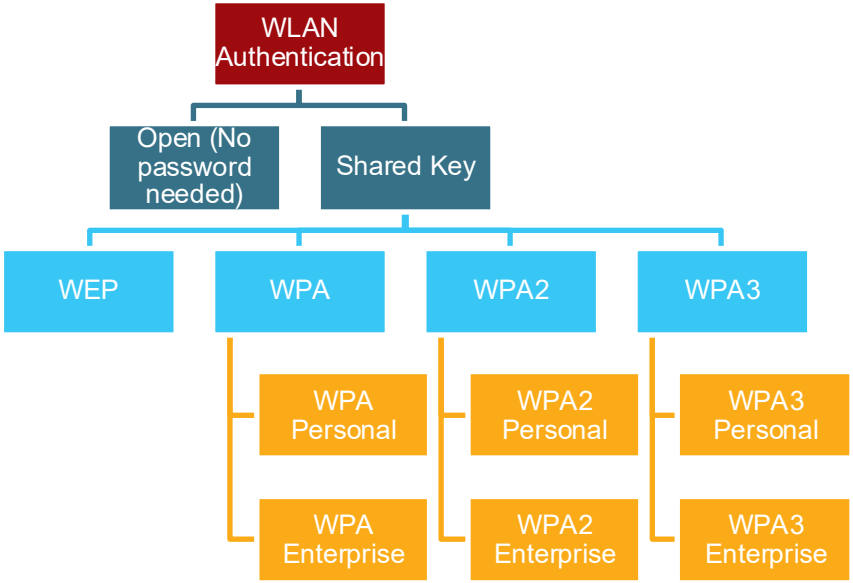
Shared Key Authentication Methods

There are currently four shared key authentication techniques available for WLANs:

- **Wired Equivalent Privacy (WEP)**
  - Early wireless security standard for encryption with support for open or shared key authentication
  - Uses RC4 stream cipher for encryption and CRC-32 checksums for integrity checking
  - Considered an obsolete security standard due to ease of key cracking – doable within 5 minutes
- **WiFi Protected Access (WPA)**
  - A Wifi Alliance standard that was an interim protocol but was considered an “acceptable” standard
  - Uses message integrity codes and introduced Temporal Key Integrity Protocol (TKIP) encryption for data confidentiality



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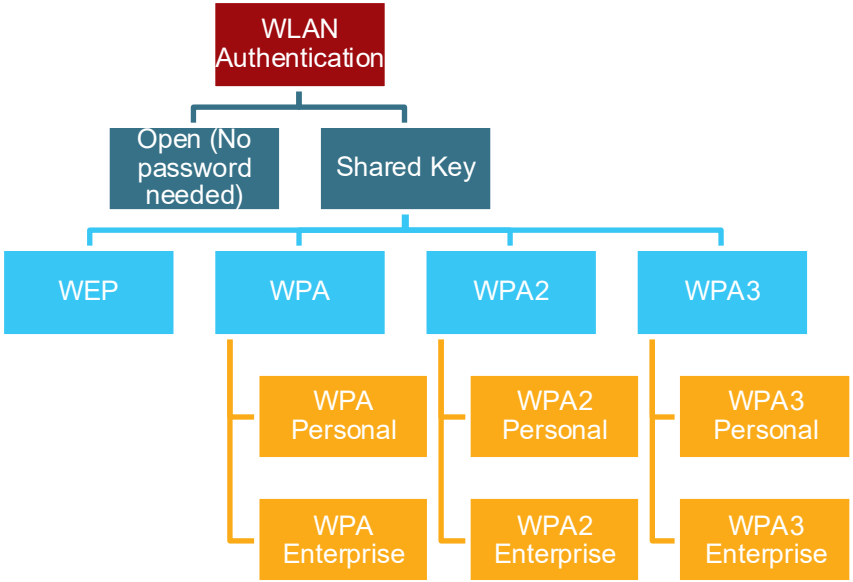
Shared Key Authentication Methods

- **802.11i / WiFi Protected Access 2 (WPA2)**
  - Was the strongest available security standard for WLANs for quite some time but is no longer considered secure now
  - Uses Counter Cipher Mode (CCMP) for integrity checking and introduced the use of Advanced Encryption Standard (AES) for encryption
- **802.11i / WiFi Protected Access 3 (WPA3)**
  - Next generation of Wi-Fi security
  - All WPA3-enabled devices use the latest security methods, disallow outdated legacy protocols, and require the use of Protected Management Frames (PMF).
  - Offers individualized data encryption, which allows data between a client and AP to be encrypted even without a password



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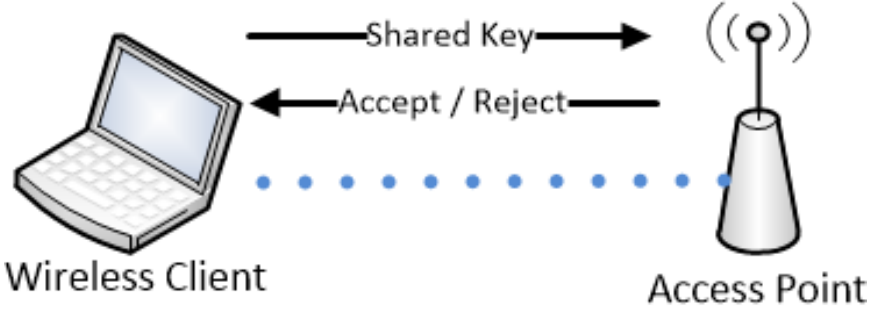
WPA Authentication Modes

WPA, WPA2 and WPA3 all offer two modes of authentication for wireless LAN clients

- **Personal Mode**
  - Uses a single pre-shared key or password between the wireless infrastructure and all WLAN users
  - Intended for home or small office networks where unauthorized access to the WLAN is not a very big risk
  - No authentication server needed



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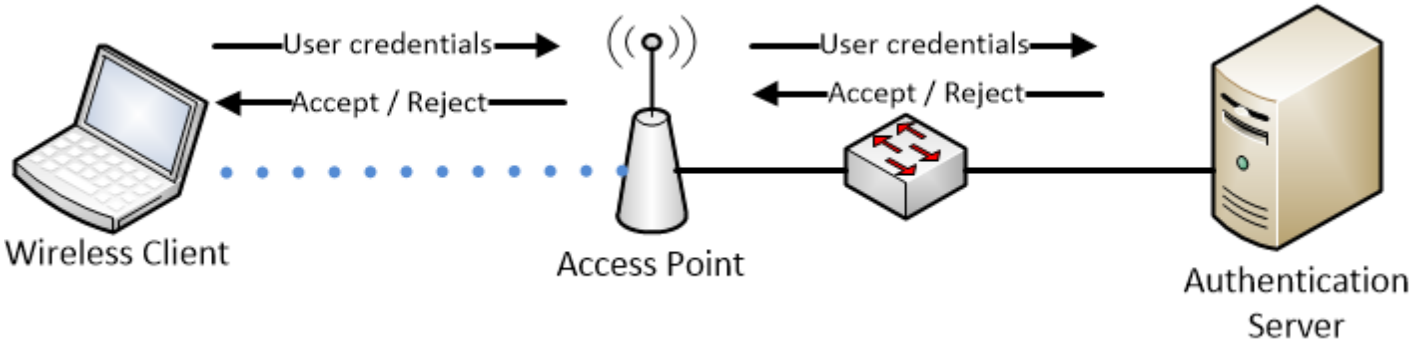
WPA Authentication Modes

WPA, WPA2 and WPA3 all offer two modes of authentication for wireless LAN clients

- **Enterprise Mode**
  - Users are assigned unique credentials that are stored in an authentication server which the AP consults to verify wireless client access authorization
  - Intended for enterprise networks with a large number of users



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5.4 Wireless LANs for SOHOs

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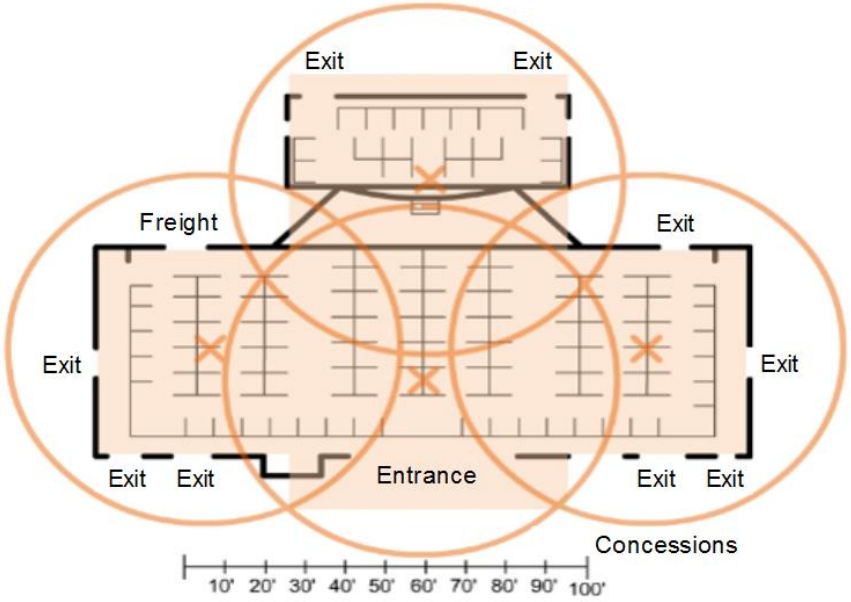
Access Point Deployment

The number of users supported by a WLAN depends on the following:

- The geographical layout of the facility
- The number of bodies and devices that can fit in a space
- The data rates users expect
- The use of non-overlapping channels by multiple APs and transmit power settings

When planning the location of APs, the approximate circular coverage area is important.



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Access Point Deployment

Some rules of thumb when installing wireless Aps or routers:

- Position APs above obstructions.
- Position APs vertically near the ceiling in the center of each coverage area, if possible.
- Position APs in /near locations where users are.
- If using several APs to cover a large area, allow coverage of APs to have a small overlap with each other so that users can roam without getting disconnected
- Important: Test speed and signal strength in areas where users are expected to be

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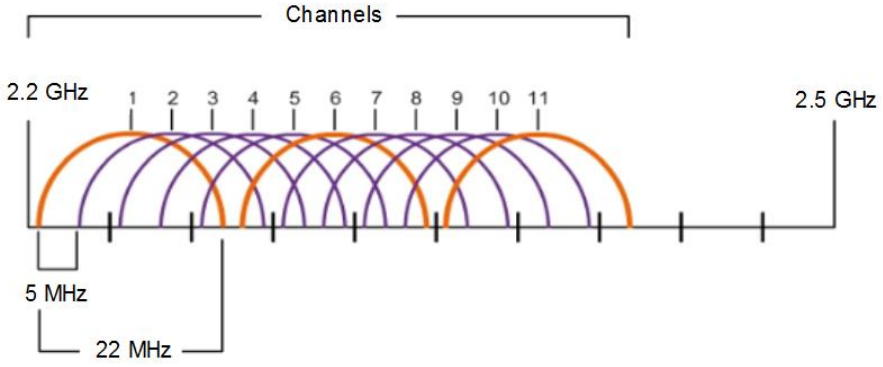
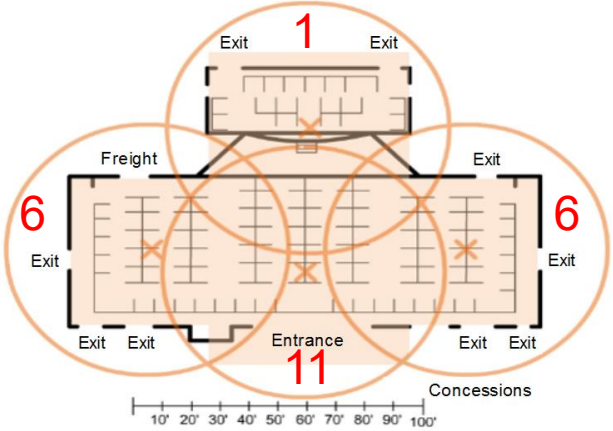
Channel Management

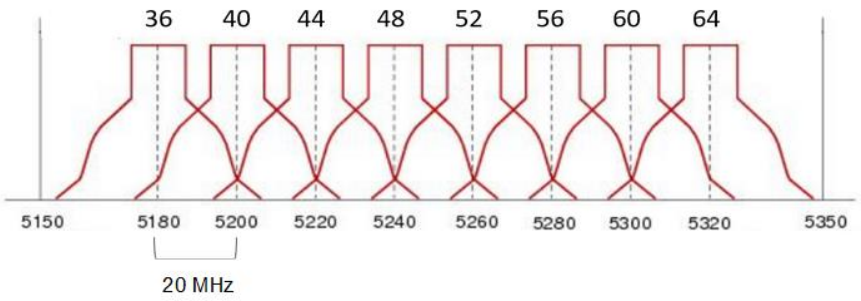
- Recall: Wireless is a shared medium with devices set to transmit and receive data on a specific range of assigned frequencies (channel)
- If the demand for a specific wireless channel is too high – i.e. too many devices on the same channel within range of each other, the channel may become oversaturated, degrading the quality of the communication.
- When deploying APs, channel selection is essential to avoid oversaturation and interference
- The 2.4 GHz band (802.11 b/g/n)
  - Subdivided into multiple channels each allotted 22 MHz bandwidth and separated from the next channel by 5 MHz.
  - Assign non-overlapping channels such as 1, 6, and 11 to APs that are in range of each other

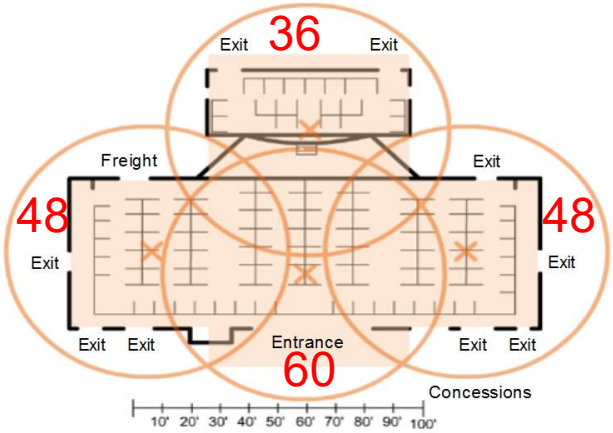




Channel Management

- Recall: Wireless is a shared medium with devices set to transmit and receive data on a specific range of assigned frequencies (channel)
- If the demand for a specific wireless channel is too high – i.e. too many devices on the same channel within range of each other, the channel may become oversaturated, degrading the quality of the communication.
- When deploying APs, channel selection is essential to avoid oversaturation and interference
- The 5 GHz band (802.11 a/n/ac)
  - Subdivided into 24 channels. Each channel is separated from the next channel by 20 MHz.
  - Assign non-overlapping channels such as 36, 48, and 60 to APs that are in range of each other





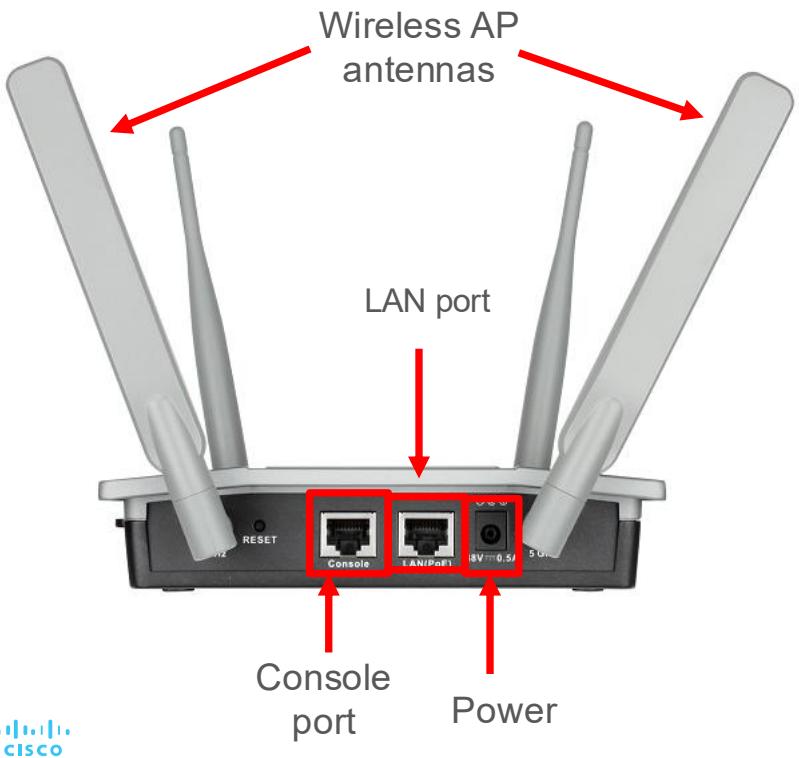
Wireless Devices for a SOHO

- In most SOHO environments, deploying autonomous access points or wireless routers using pre-shared keys is sufficient to address the needs of a wireless network
- The differences:

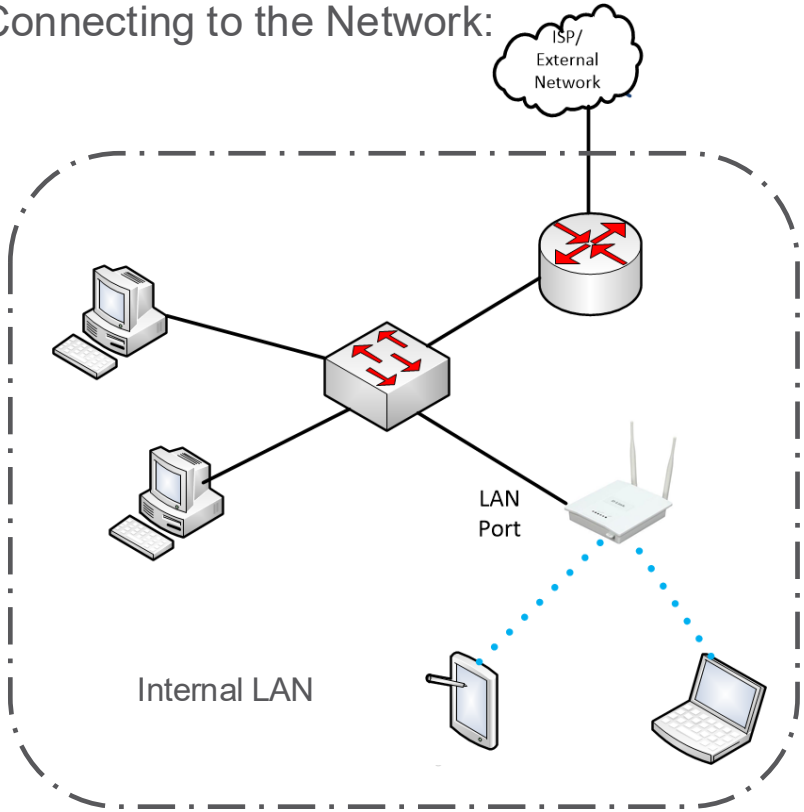
	Access Point	Wireless Router
Connection types supported	Connects users to the LAN using a wireless connection.	Usually, can connect users through either wireless or a limited number of wired connections
Packet forwarding	Exchange traffic within the same subnet only. Separate router needed to reach other networks	Can route traffic between networks
Available Services	Does not provide network services	Provides common network services needed in a LAN. e.g. DHCP, DNS, packet filtering
When to Use	Best used in LANs where an existing infrastructure including a network gateway is already available.	Best used in LANs with only a few users (e.g. < 30) where an all-in-one device can serve as the entire infrastructure

Connecting a Wireless AP

- A wireless AP acts like a hub for wireless clients



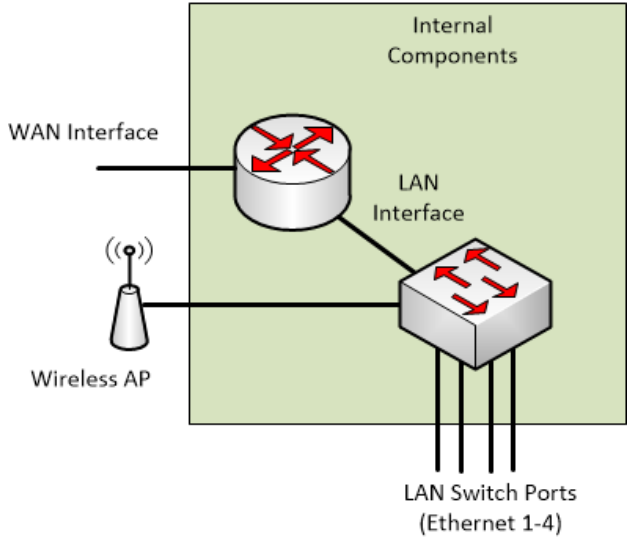
Connecting to the Network:

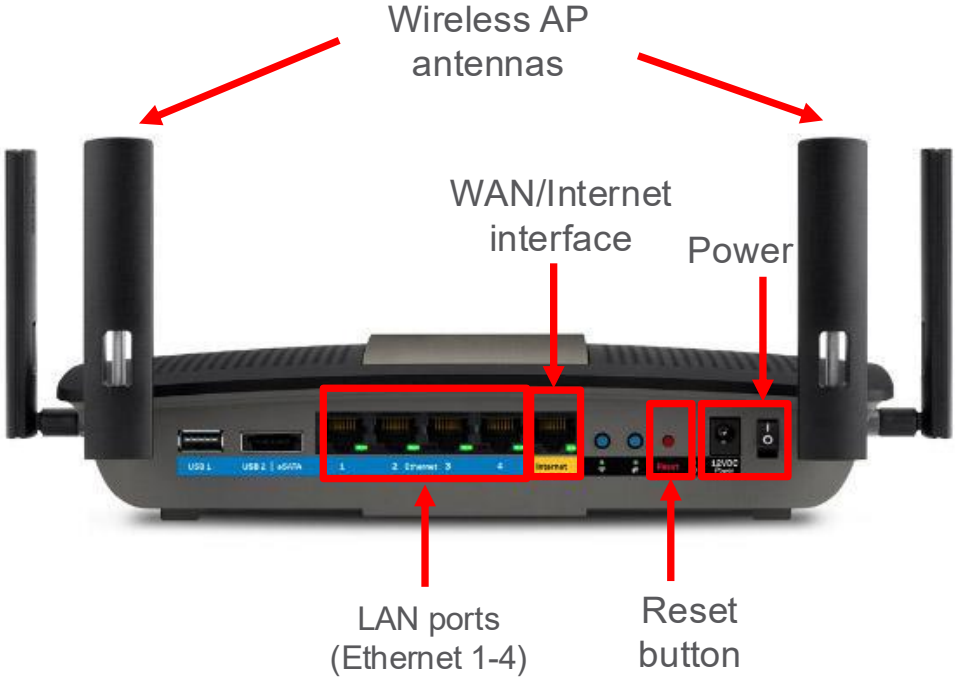


Connecting a Wireless Router

- Recall: A wireless router is often designed as an all-in-one device that internally contains a router, access point and switch.

- Equivalent Components:





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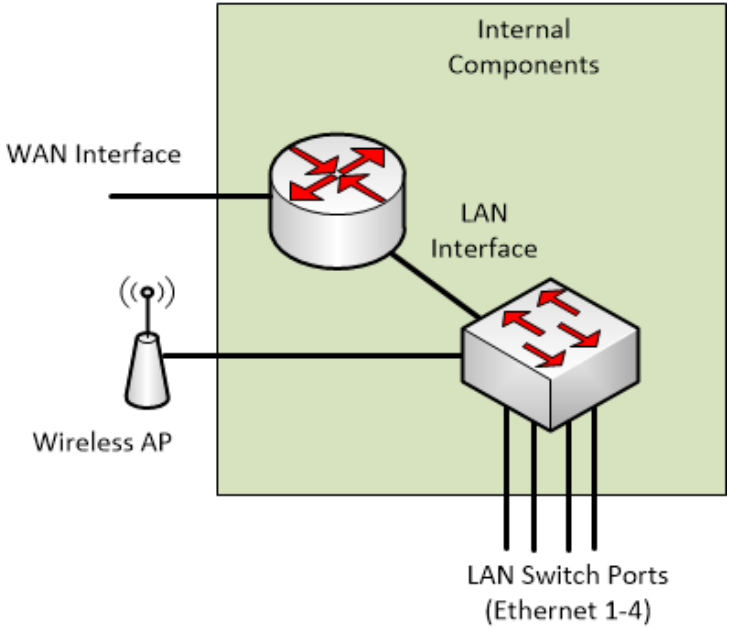
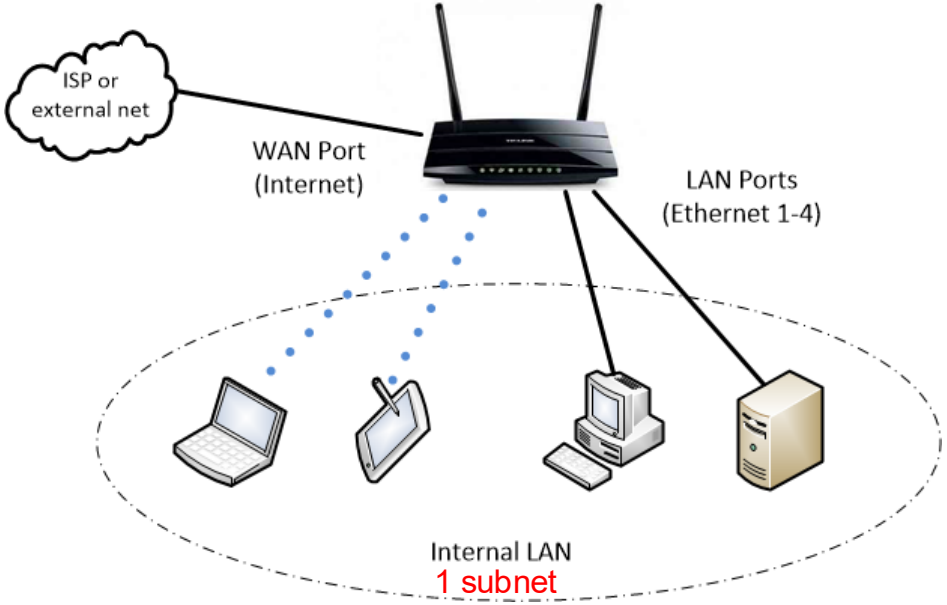
Connecting a Wireless Router

Equivalent Components:



Connecting to the Network:



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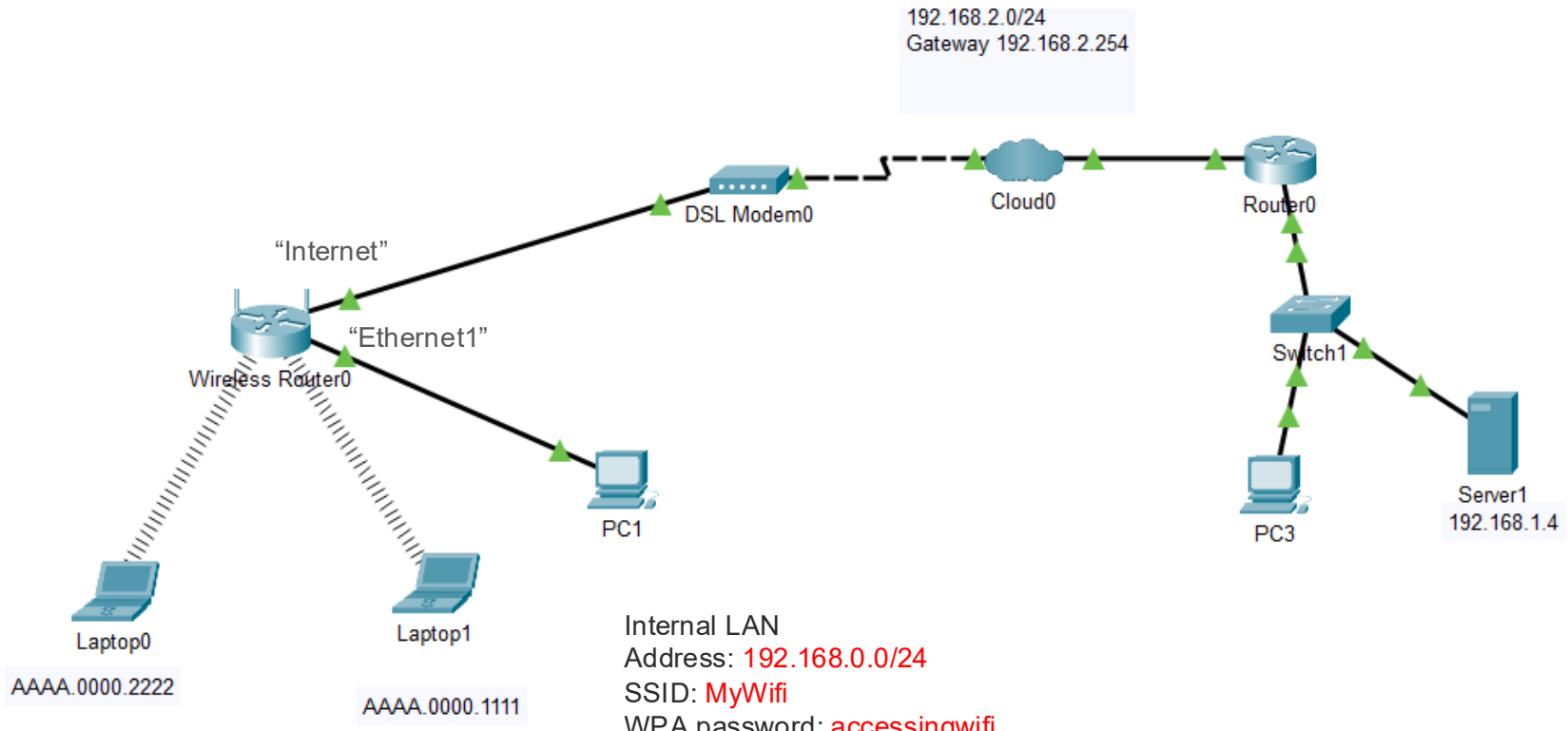
Sample Topology



Internal LAN
Address: 192.168.0.0/24
SSID: MyWifi
WPA password: accessingwifi

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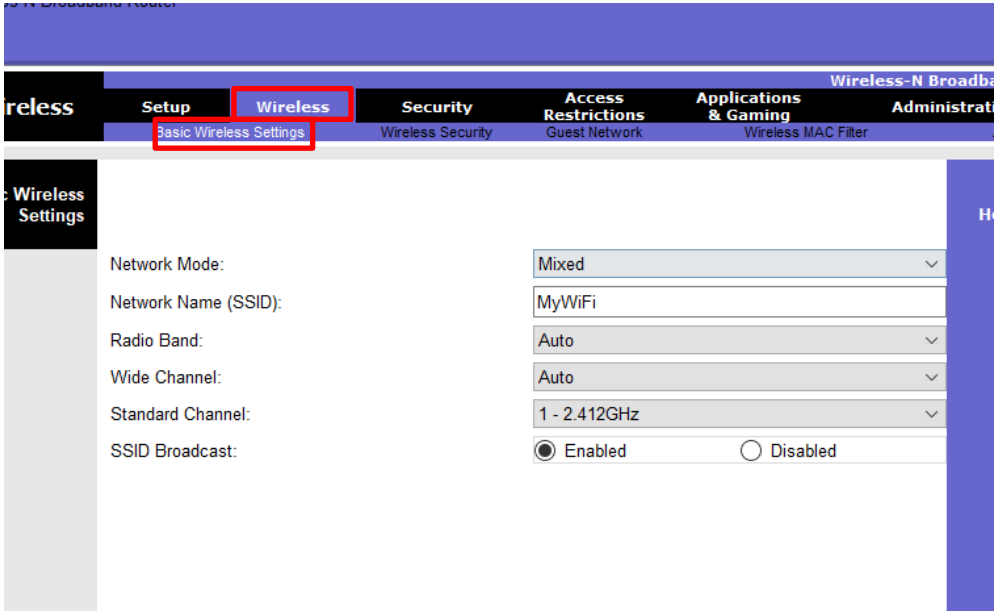
Wireless Settings

The following the are required parameters when configuring an AP or wireless router for WLAN operations



- **Network Mode**
  - Sets which standard to use (802.11 a/b/g/n/ac)
  - Note: If supporting backward compatibility, connecting a client with a lower speed standard can slow down overall network performance
- **SSID**
  - A unique identifier for a particular WLAN
  - A WLAN client must know the WLAN SSID to establish a connection
- **SSID Broadcast**
  - Setting this to ‘On’ allows the AP or wireless router to announce the presence of the WLAN
  - Setting this to ‘Off’ hides the WLAN and requires clients to be preset with the WLAN SSID beforehand

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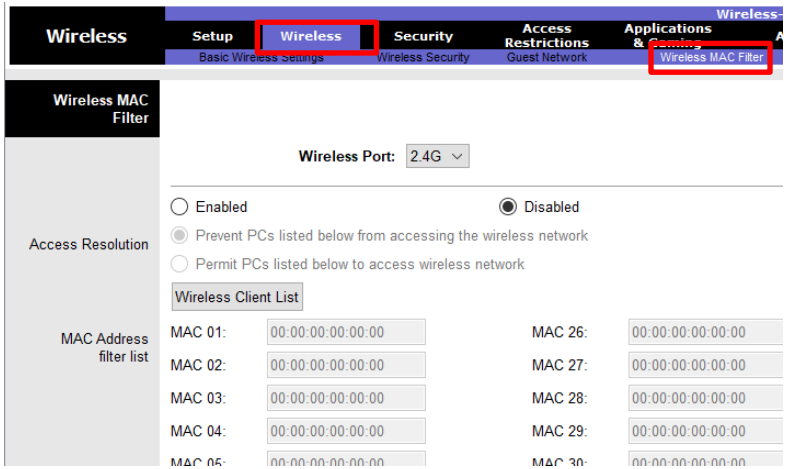
Wireless Security Settings

The following are WLAN security parameters that can be configured on an AP or wireless router

- **Security Mode**
  - Set to WEP, WPA or WPA2.
  - Some devices allow selection of encryption. Select between TKIP or AES
- **If using 'Personal Mode' or 'PSK'**
  - Password - Refers to the common password shared among all devices of the WLAN
- **MAC Filtering**
  - Set a blacklist or whitelist of MAC addresses to control hosts allowed to connect to the WLAN



Address Settings

Important: Network Address Translation is often enabled between the Internet and LAN interfaces on a wireless router.

Because of this:

- All internal LAN hosts are collectively identified using the router's Internet address when they exit to the WAN

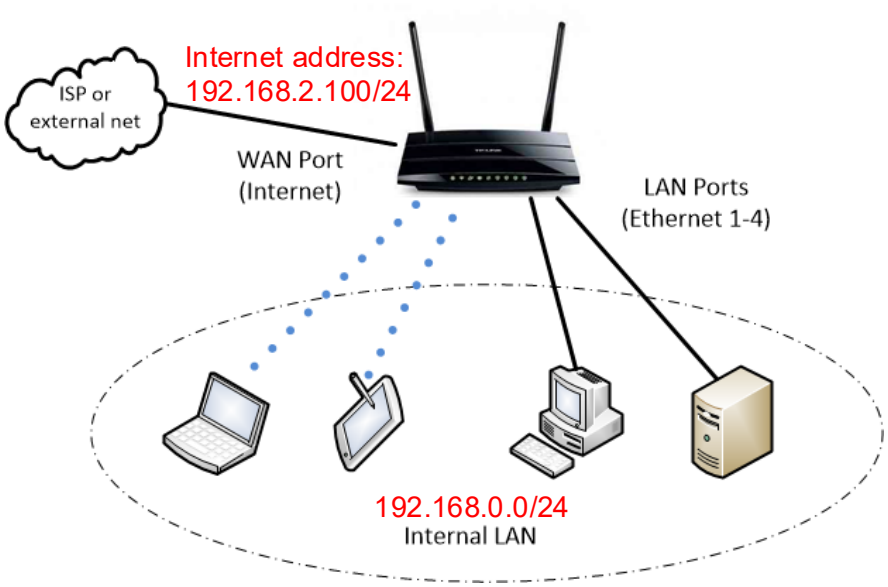
  i.e. The PC / server / mobile devices are all known as 200.200.200.100 to everybody on ISP or external net

- All hosts on the internal LAN can freely access any host on the WAN

  i.e. The PC/server/mobile devices can ping any host in the ISP or external net

- Hosts on the WAN side cannot directly access any host in the internal LAN

  i.e. Any host on the ISP or external net cannot ping the PC/server/mobile devices



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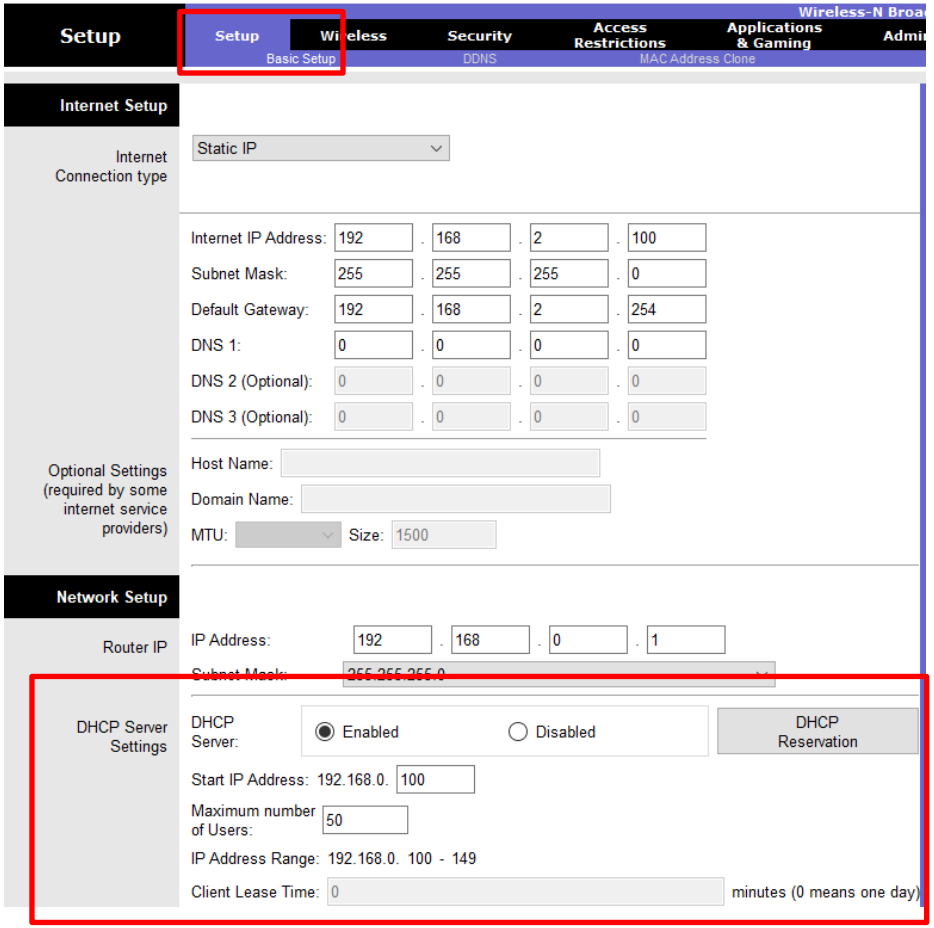
DHCP Settings

For wireless routers, a DHCP pool can optionally be configured to automatically assign addresses to the LAN side clients connected to the built-in switch and wireless LAN

- **DHCP Server** - set to enabled
- **Start IP address -** Provide the starting IP address to be leased from the pool
- **Maximum Users –** Set to the number of addresses in the pool



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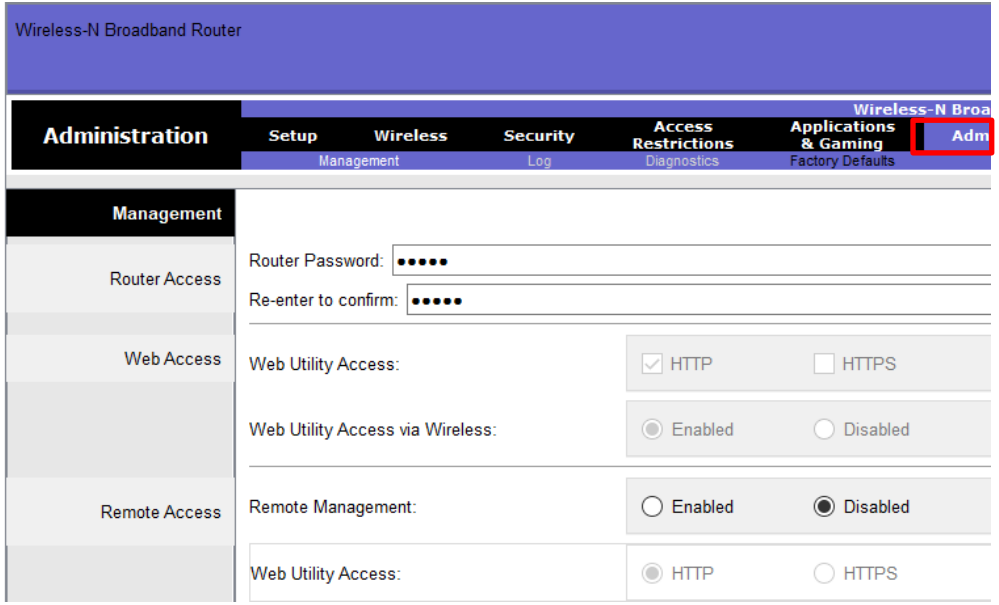
Administrative Settings

The following the parameters that can be additionally configured

- **Router password**
  - Secured the web UI with a password
  - By default – password is ‘admin’
- **Remote management**
  - Enables/disables web UI access to hosts from a different subnet



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5.5 Wireless LANs for Enterprises

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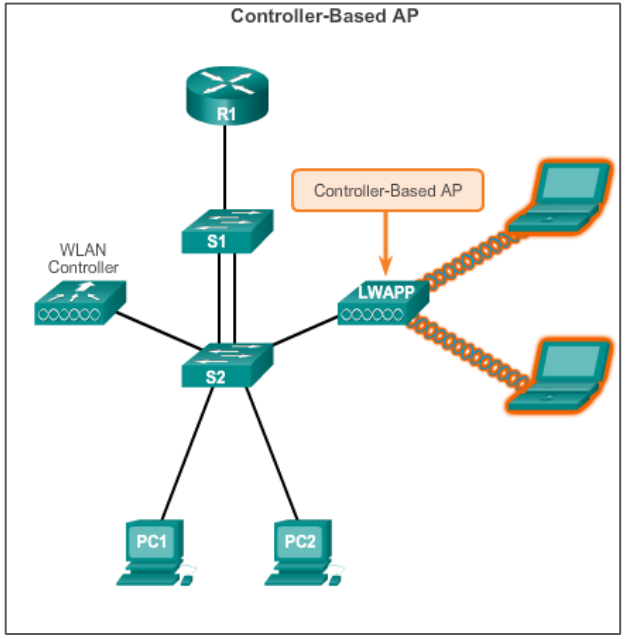
Enterprise WLAN Characteristics

- A wireless network for a medium to large organization introduces the following complexities:
  - Multiple access points to provide adequate coverage for the users
  - Access points need to have consistent WLAN settings across the facility
  - Users belonging to different groups/networks connecting to the WLAN
  - High risk of unauthorized WLAN access
- Enterprise WLANs are commonly designed with the following elements:
  - Controller-based / lightweight access points which can support VLAN tagging installed throughout the facility
  - Wireless controllers (WLC) managing AP configuration and monitoring
  - Unique WLAN login per user using WPA enterprise mode security



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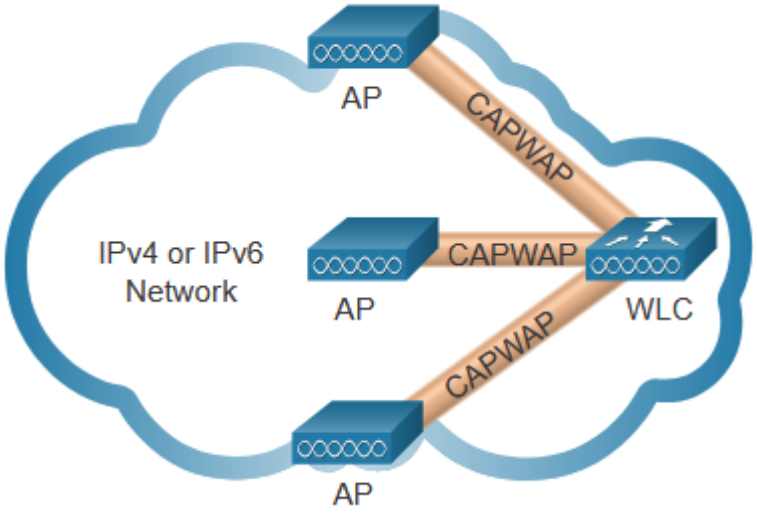
Introduction to CAPWAP

- Control And Provisioning of Wireless Access Points (CAPWAP) is an IEEE standard protocol that enables a WLC to manage multiple APs and WLANs.
- Reduces the administrative overhead of managing and monitoring a large number of APs in a network
- Encapsulates and forwards WLAN client traffic between an AP and a WLC over tunnels using UDP ports 5246 and 5247.
- Capable of encrypting the transmission of control and management traffic between AP and WLC using Datagram Transport Layer Security (DLTS).



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CAPWAP Split MAC Architecture

The CAPWAP split MAC concept does all the functions normally performed by individual APs and distributes them between two functional components:

- Access Point MAC Functions
- Wireless Controller

Access Point MAC Functions	Wireless Controller MAC Functions
Beacons and probe responses	Authentication
Packet acknowledgements and retransmissions	Association and re-association of roaming clients
Frame queueing and packet prioritization	Frame translation to other protocols
MAC layer data encryption and decryption	Termination of 802.11 traffic on a wired interface

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Wireless LAN Controllers

Wireless LAN Controllers have Layer 2 switch ports and virtual interfaces that are created in software and are very similar to VLAN interfaces.

- Each physical port can support many APs and WLANs.
- The ports on the WLC are essentially trunk ports that can carry traffic from multiple VLANs to a switch for distribution to multiple APs.
- Each AP can support multiple WLANs.

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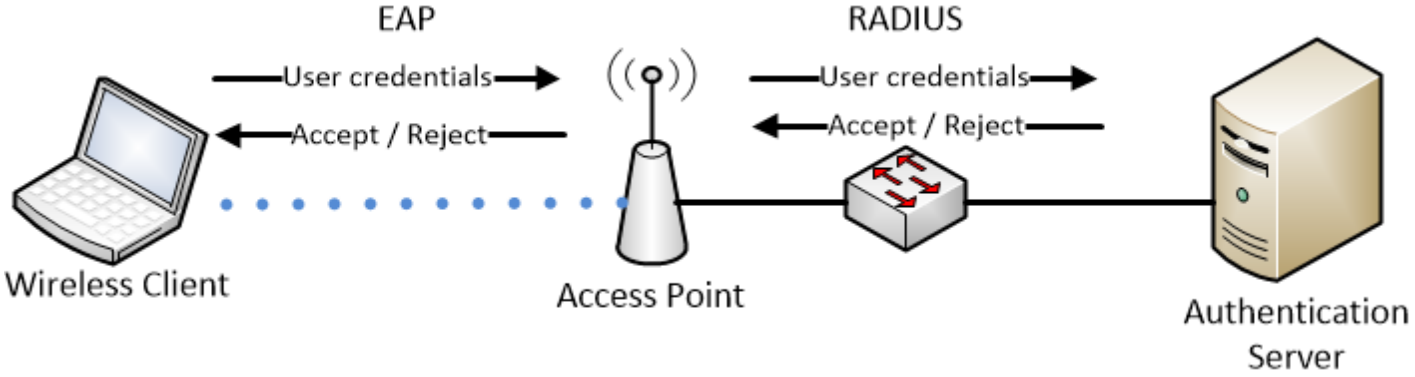
Enterprise Mode WPA Security

- WPA, WPA2 and WPA3 enterprise mode security lowers the risk of unauthorized access due to a password leak or brute forcing
  - Each user is assigned unique login credentials for the WLAN
  - Clients use the 802.1X standard with Extensible Authentication Protocol (EAP) to communicate user credentials with the AP as required by the AAA server
  - AP or WLC uses an AAA protocol (usually RADIUS) to relay user credentials to AAA server
  - AAA server performs login authentication, authorization and accounting when users connect to the WLAN



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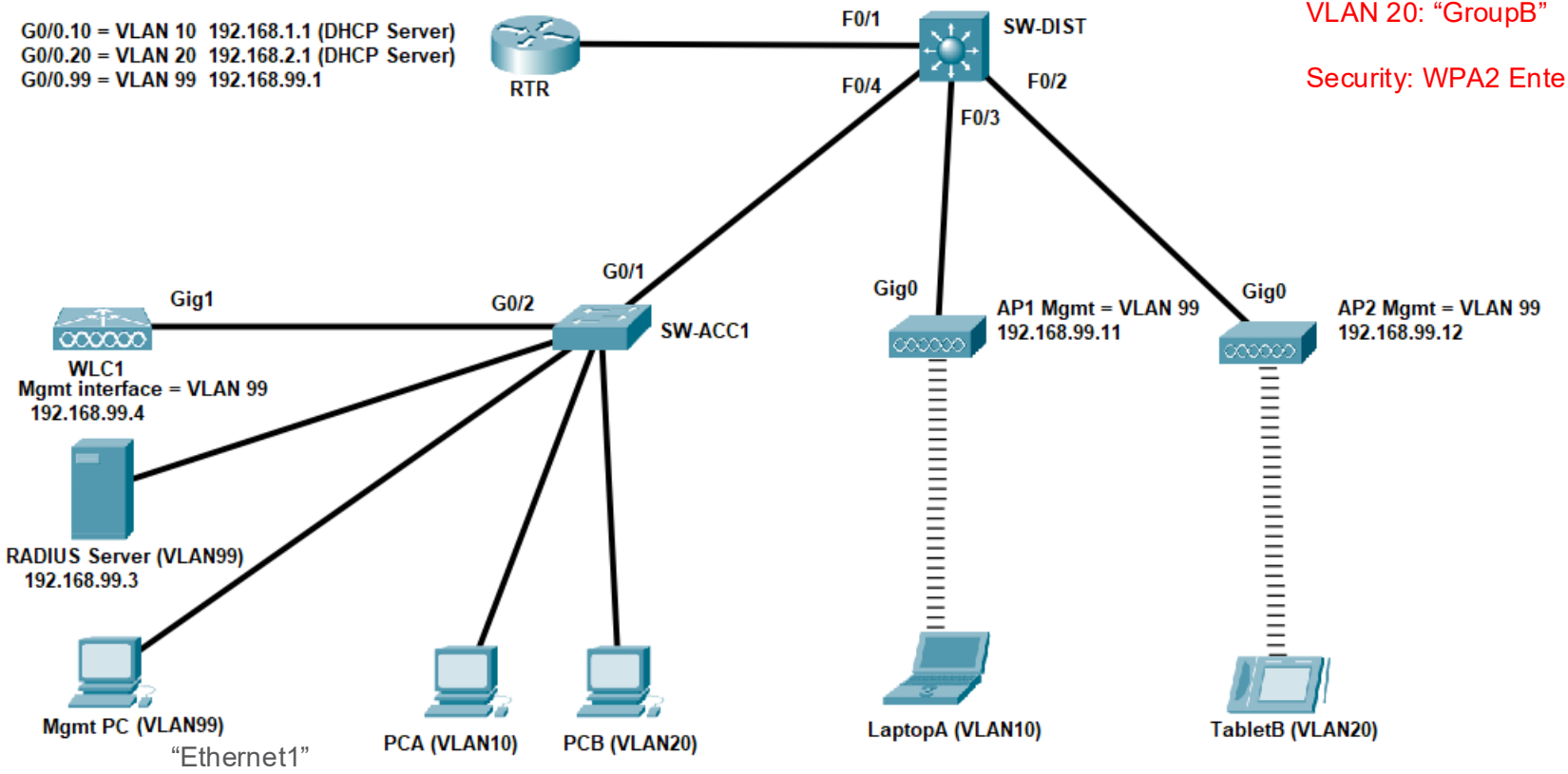
Sample Topology

WLAN SSIDs
VLAN 10: “GroupA”
VLAN 20: “GroupB”

Security: WPA2 Enterprise



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General Configuration Tasks

- To set up an enterprise WLAN using lightweight APs and enterprise mode security on a network that has existing VLANs, the following tasks need to be done

1. Prepare switches ports for connection of the WLC and APs
2. If using enterprise security mode, configure an AAA server with user accounts
3. Configure WLC with basic and WLAN settings
4. Configure access points

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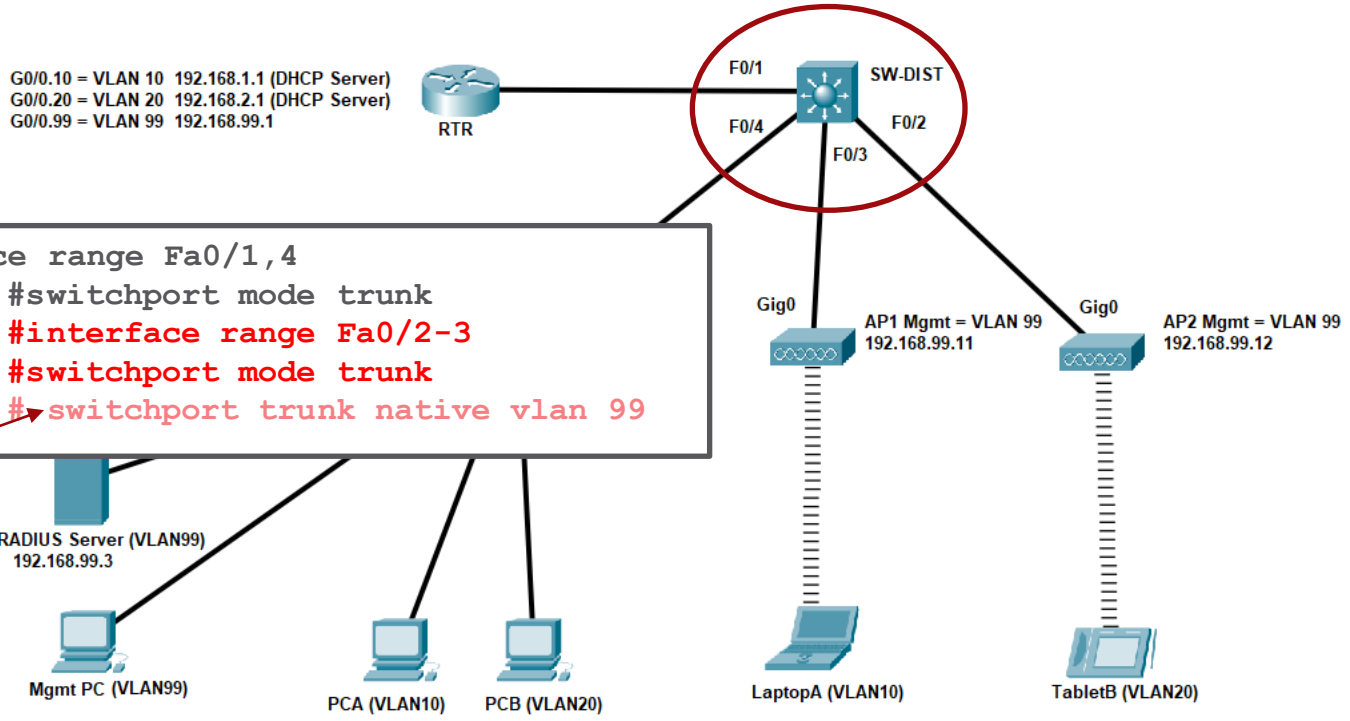
Switch Configuration

- Since APs and WLCs will be supporting traffic from multiple VLANs, the switch ports connected to these devices must be set to trunk mode

```
SW-DIST(config)#interface range Fa0/1,4
SW-DIST(config-if-range)#switchport mode trunk
SW-DIST(config-if-range)#interface range Fa0/2-3
SW-DIST(config-if-range)#switchport mode trunk
SW-DIST(config-if-range)#switchport trunk native vlan 99
```

Needed only if using PT due to a bug allowing WLC and AP addresses to be set only on untagged VLANs



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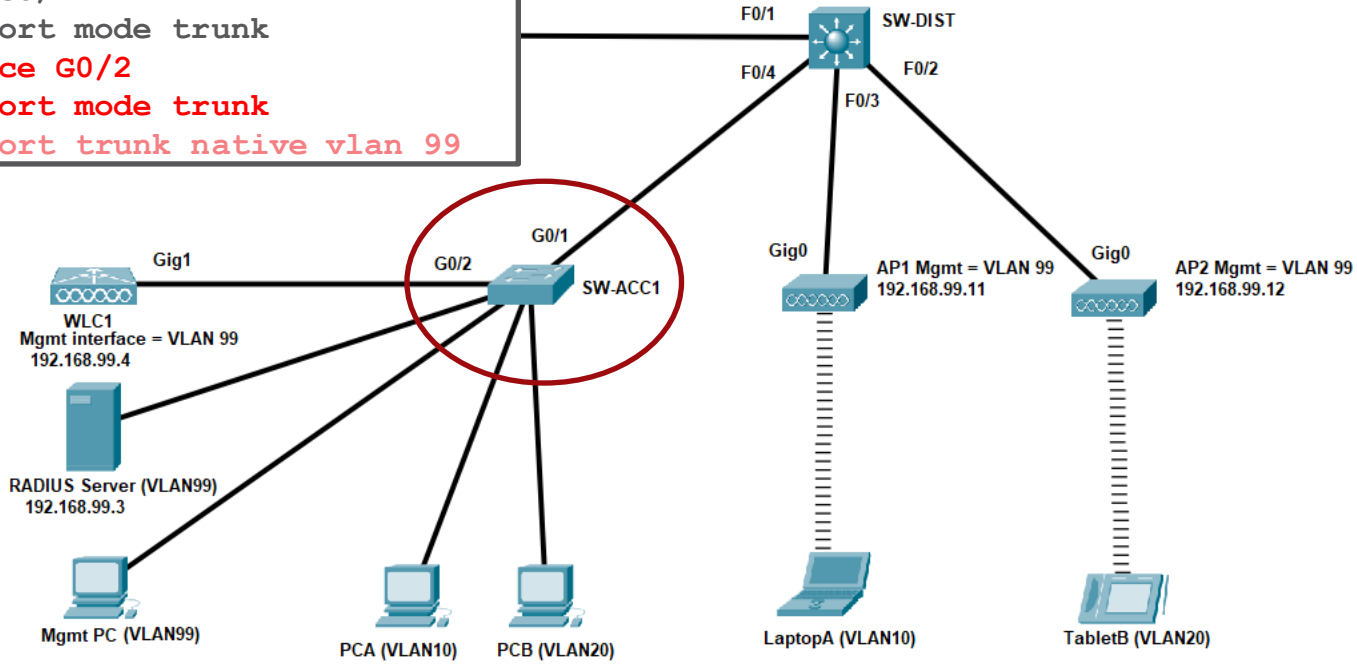
Switch Configuration

- Since APs and WLCs will be supporting traffic from multiple VLANs, the switch ports connected to these devices must be set to trunk mode

```
SW-ACC1(config)#interface G0/1
SW-ACC1(config-if)#switchport mode trunk
SW-ACC1(config-if)#interface G0/2
SW-ACC1(config-if)#switchport mode trunk
SW-ACC1(config-if)#switchport trunk native vlan 99
```

Needed only if using PT due to a bug allowing WLC and AP addresses to be set only on untagged VLANs



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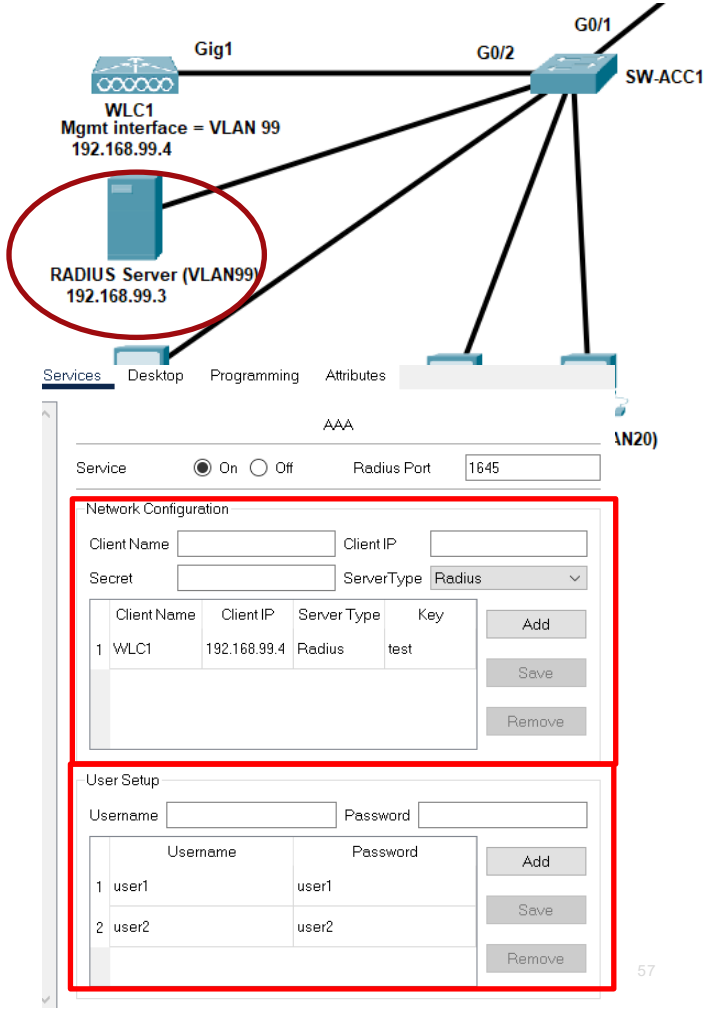
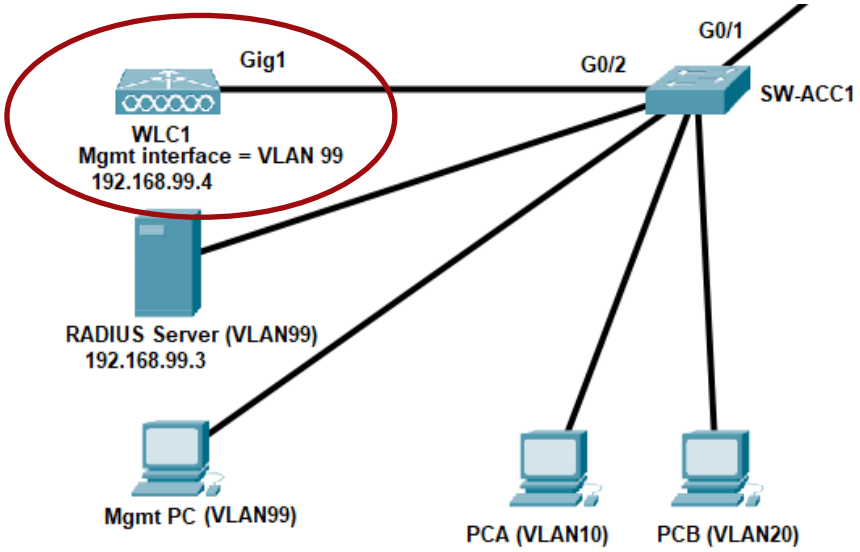
AAA Server Configuration

- RADIUS is commonly used for the AAA server. The server must be configured with user accounts and with RADIUS client information so that it recognizes the WLC which will contact it to authenticate users
- RADIUS Client configuration
  - Client Name – any name to identify the WLC
  - Client IP – IP address of the WLC
  - Secret - Refers to the common password to be shared between server and WLC
- User setup – create usernames and passwords for WLAN users
- Note: Setting up a real RADIUS server is out of the scope of our course



WLC Configuration Tasks

- Configuration tasks on the WLC:
  1. If using enterprise security mode, configure AAA server details
  2. Create a dynamic interface per WLAN to be hosted
  3. For each WLAN, configure wireless and security settings
- Configuration of the WLC may be performed by accessing its HTTP/S Web GUI using its management IP address – use browser of a PC



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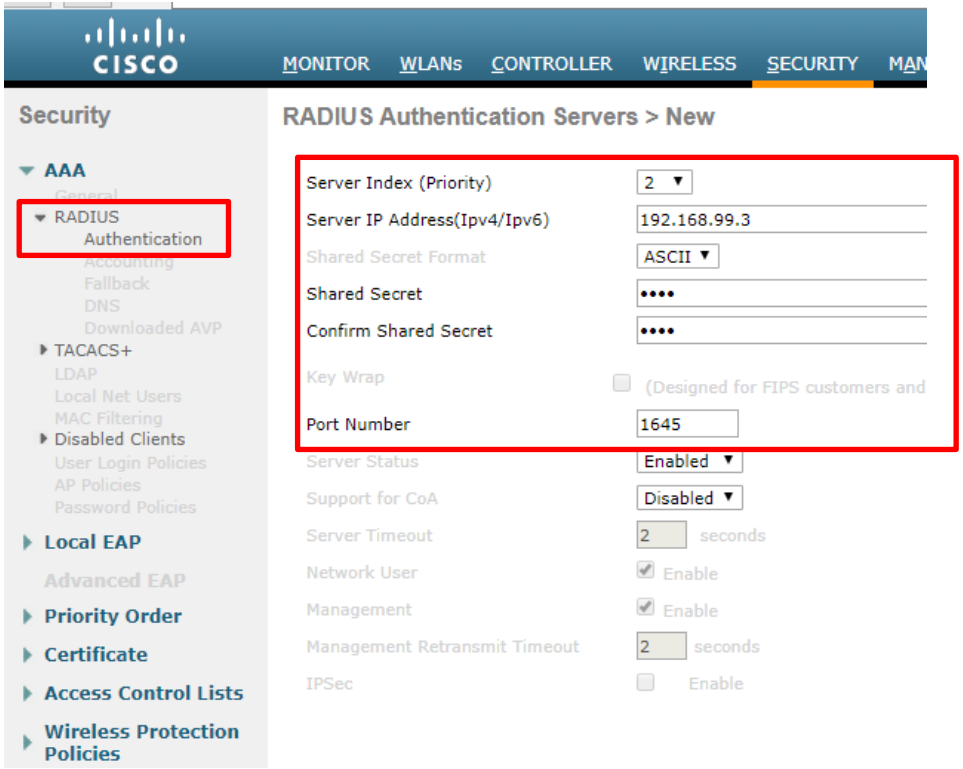
WLC Configuration – Define AAA Server

- Enterprise security mode choice requires an the wireless AP/ router / WLC (as in the sample scenario) to work with an AAA server for user authentication
- Three pieces of information are required:
  - **RADIUS server IP address** – IP address of the server.
  - **UDP port numbers** –UDP ports 1812 for RADIUS Authentication, and 1813 for RADIUS Accounting, but can also operate using UDP ports 1645 and 1646.
  - **Shared key** – common password shared between the AP / wireless router / WLC and an authentication. This should match the configured password associated on the server



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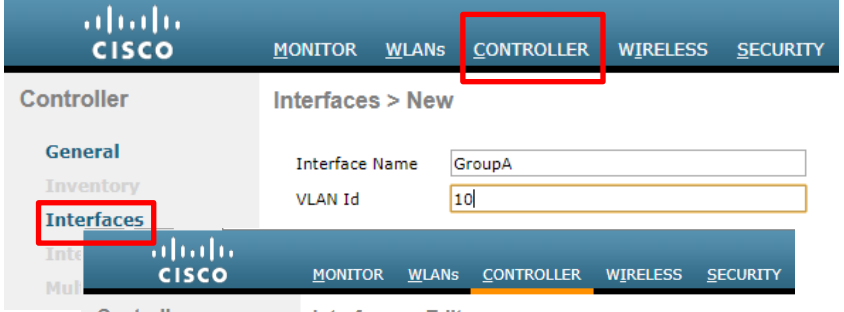
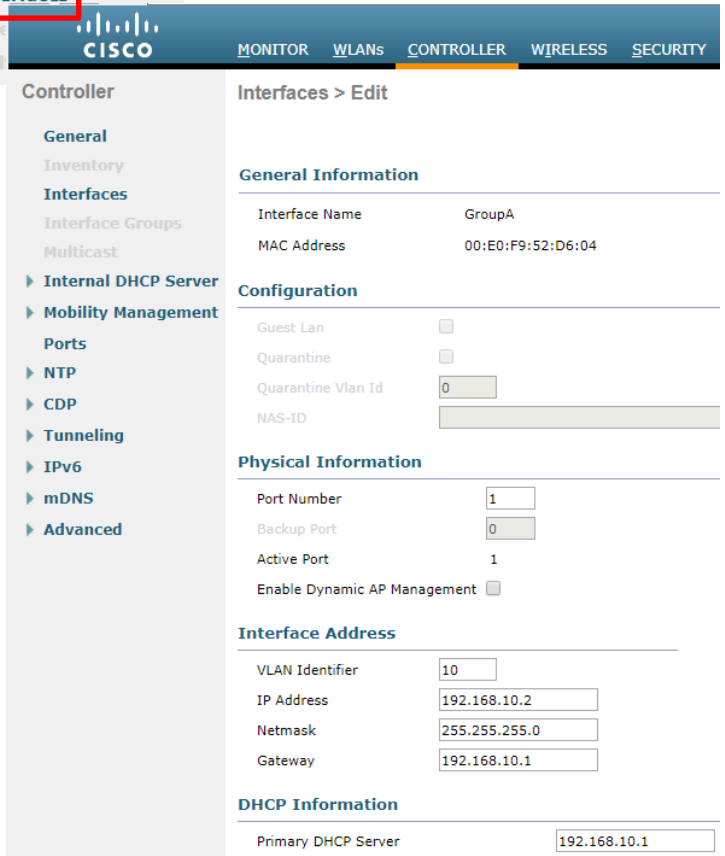
WLC Configuration - Interfaces

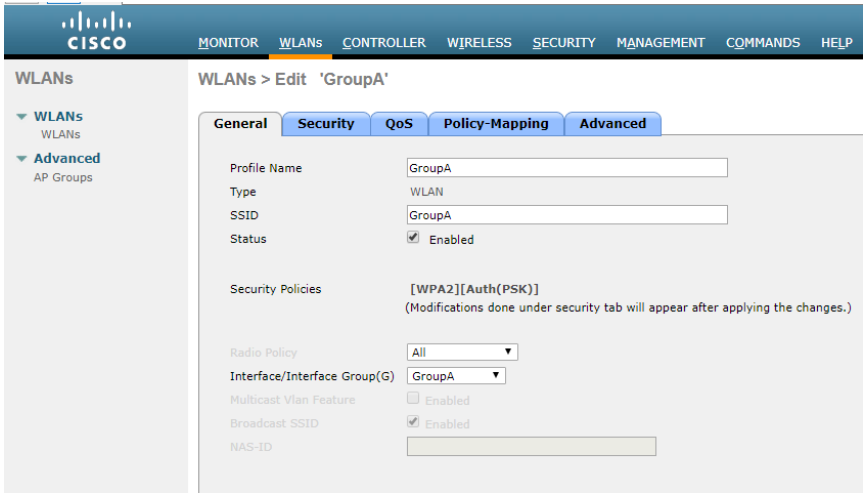
- For each WLAN that the network will host, a dynamic interface needs to be created
  - Interface Name: Any string to identify the interface
  - VLAN ID: VLAN number assigned to the user group that will use the WLAN
  - Port number: physical port of the WLC connected to the network infrastructure
  - IP address / gateway : IP settings of the dynamic interface – must come from the subnet assigned to the VLAN
  - Primary DHCP: DHCP server address hosting the address pool for the VLAN if it uses DHCP for host addressing





WLC Configuration - WLANs

- Multiple WLANs may be created on the WLC. The following settings are basic wireless settings needed for each:
  - Profile Name: Any string to identify the WLAN configuration profile
  - SSID: network name
  - Status: Enabled/Disable to turn the WLAN on/off
  - Interface: WLC dynamic interface associated with the subnet/VLAN where the wireless clients connecting to this WLAN will be assigned



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WLC Configuration – WLAN Security

The following settings are wireless security settings needed for each WLAN:

- Layer 2 Security: select the preferred security protocol
- If choosing WPA/WPA2:
  - Select which will be used (WPA Policy or WPA2 Policy)
  - Select encryption algorithm (AES or TKIP)
- Select authentication key management: enterprise mode (802.1X) or Personal mode (PSK)
  - For PSK: supply the shared password
  - For 802.1X: Use the AAA Servers tab to select the authentication server

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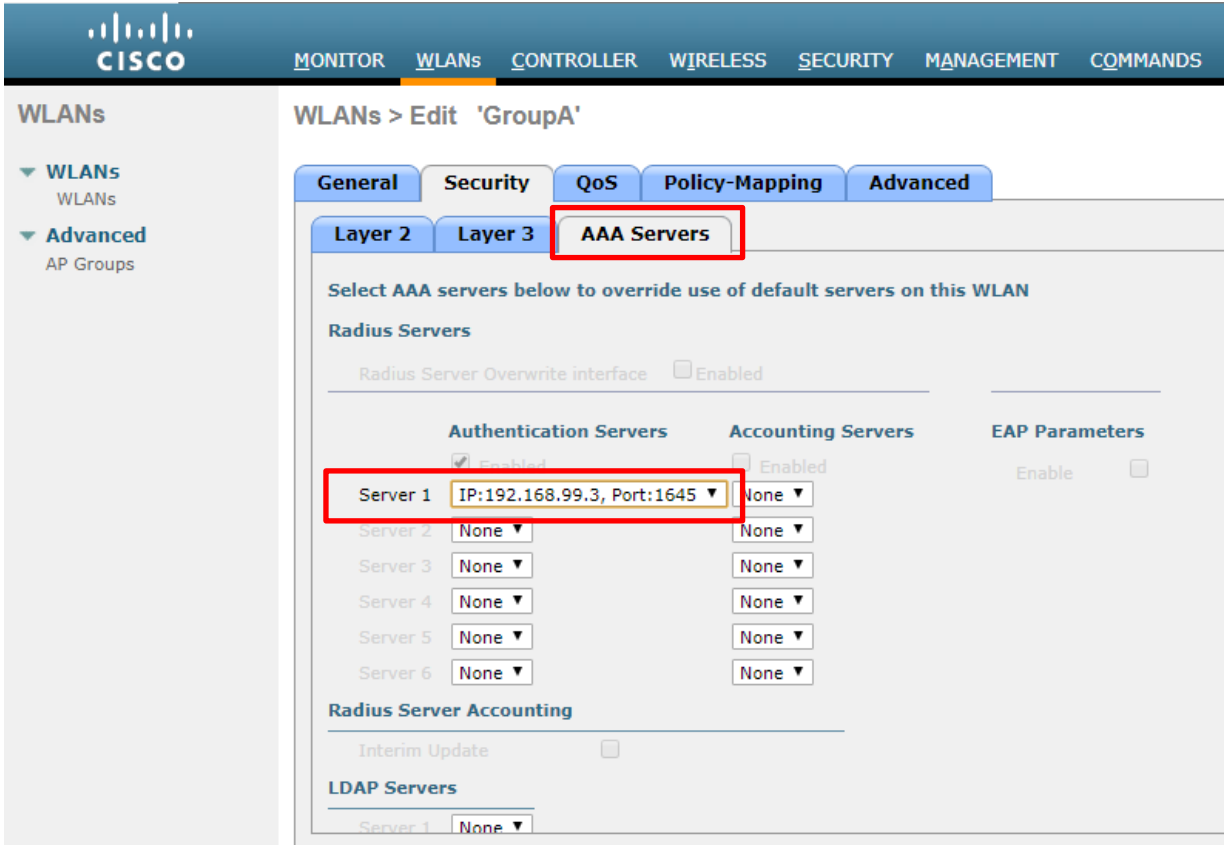
WLC Configuration – 802.1X AAA Servers

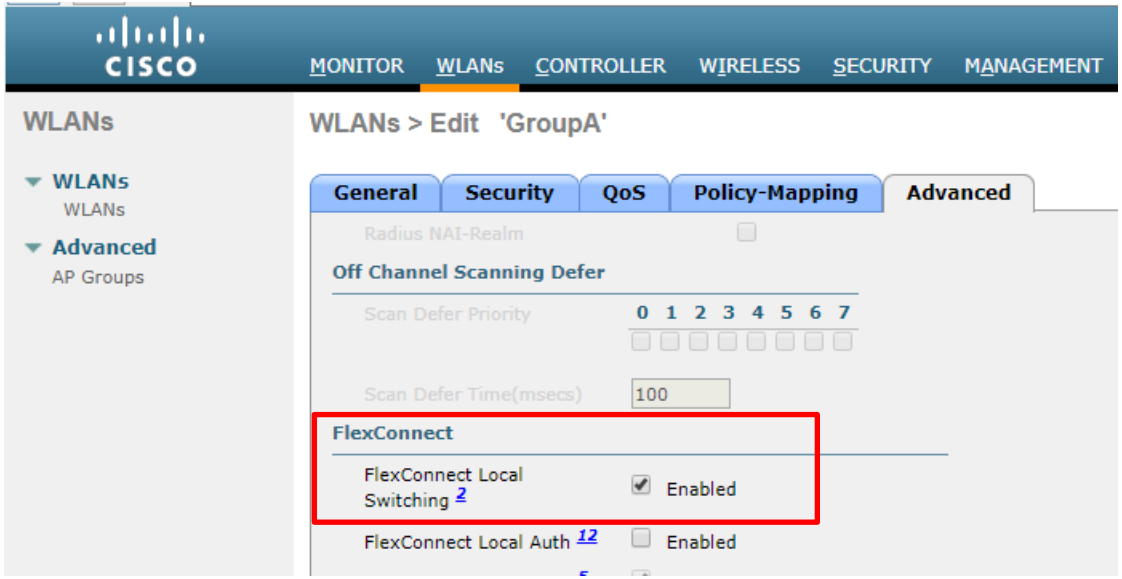
If enterprise mode security is used (802.1X), additionally select which configured AAA server the WLAN will reference for user authentication

AAA server must be preregistered on the WLC



WLC Configuration – Advanced

To allow the WLC to automatically configure the APs to associate each WLAN to an assigned VLAN associations, enable the FlexConnect Local Switching option (works if using Cisco access points that support this feature)



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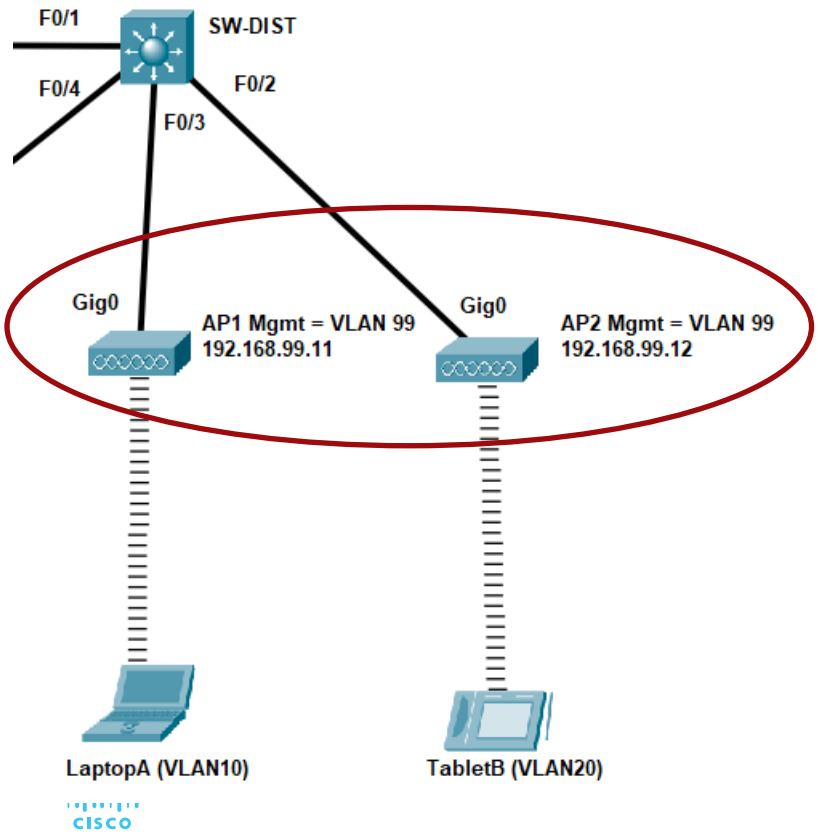
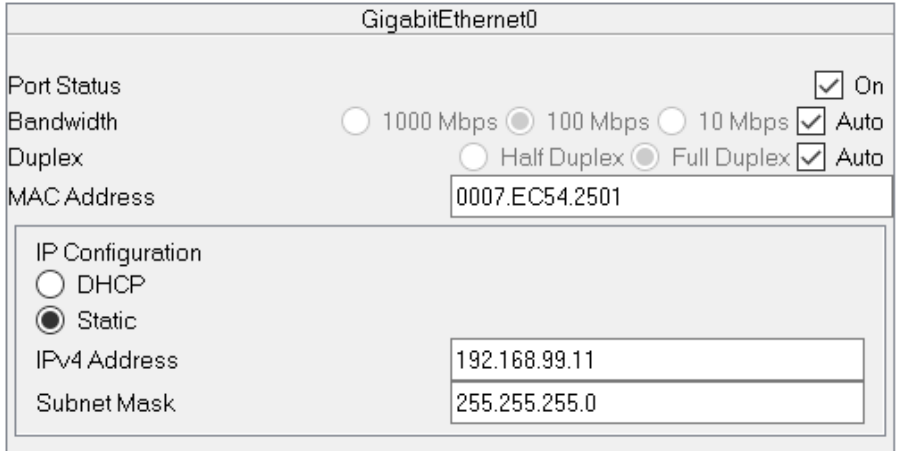
Lightweight AP Configuration



When setting up lightweight APs, the following are key configurations:

1. Assign an IP address and default gateway either statically or using DHCP to allow them to communicate with the network WLC
2. Cisco APs will automatically be discovered by the WLC and configured to make the configured WLANs available to wireless clients



What did you learn in this module?

- Wireless technology uses the unlicensed radio spectrum to send and receive data.
- A Wireless LANs (WLANs) is a wireless network spanning a single organization site and is commonly based on the IEEE 802.11 standard
- WLAN networks operate in the 2.4 GHz and 5 GHz frequency band and are capable of different maximum bandwidths depending on which variant of IEEE 802.11 is used.
- WLAN devices include:
  - Wireless NICs used to connect hosts to a wireless network
  - Access points used to provide wireless connectivity to hosts
  - Wireless routers which provide the functionalities of router, switch and access point in 1 device
  - Wireless controllers which can centrally manage multiple APs in a large network
- 802.11-based WLANs use CSMA-CA for media access control and use SSIDs to identify WLANs

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What did you learn in this module?

- To connect to an 802.11 WLAN, wireless hosts undergo a 3-step process: Discovery, Authentication, Association before being able to communicate on the wireless network
- Wireless networks are susceptible to threats, including: data interception, wireless intruders, DoS attacks, and rogue APs.
- To keep wireless intruders out and protect data, WLAN may be protected using
  - Early security features : SSID cloaking and MAC address filtering.
  - Shared key authentication techniques : WEP, WPA, WPA2, and WPA3.
- WPA/WPA2/WPA3 support 2 mode of authentication
  - Personal mode – single common password for all users, authentication done by AP/router/WLC
  - Enterprise mode – individual account per user, authentication done by dedicated AAA server

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What did you learn in this module?

- To set up a SOHO WLAN, wireless routers or autonomous APs are often used and configured with SSID, shared key authentication technique, shared key string that must match with those configured on the wireless clients
- To set up an enterprise WLAN, a WLC is used to manage multiple APs and enterprise mode WPA/WPA2/WPA3 is used to coordinate with an AAA server for authentication. The following are configurations needed:
  - Prepare switch interfaces for WLCs and APs as trunks
  - Set up AAA server with user accounts and WLC identity details
  - Configure WLC with a dynamic interface, VLAN ID, SSID, security settings per WLAN
  - Configure APs with management IP addresses to allow them to receive centralized control from the WLC

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